A ROBUST SHIFTED SCHUR-COMPLEMENT PRECONDITIONER FOR THE IMPLICIT LINEARISED BOUSSINESQ EQUATIONS*

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Abstract. Implicit time integration of the incompressible Boussinesq equations removes the gravity-wave CFL restriction that limits explicit ocean models, at the cost of solving a saddle-point system coupling velocity, buoyancy and pressure at every step. On the thin domains characteristic of ocean basins this system is hard to precondition: the $H(\mathrm{div})$ velocity block is poorly conditioned on anisotropic meshes, and the gravity-wave coupling makes the pressure Schur complement sensitive to the time step. We propose and analyse a preconditioner built from a shifted problem in which the pressure equation is augmented by a small multiple δ of the pressure mass matrix. The shift makes the pressure block invertible in isolation, permitting an exact Schur-complement elimination of the pressure and reducing each solve to a velocity–buoyancy block with a grad–div augmentation, treated by multigrid with a vertical column-patch smoother. Our main result bounds the spectral distance between the shifted and unshifted operators explicitly in terms of δ, the time step, the buoyancy frequency and the aspect ratio, in a norm weighted by the stratification operator; bounds are given with and without the assumption that the velocity block is symmetric, and they agree to leading order. The inf–sup constant is shown to be exactly independent of the aspect ratio throughout the thin-domain regime, so the preconditioner is uniform precisely in the limit where standard approaches degrade, and the analysis predicts a two-regime scaling for δ whose cross-over is governed jointly by the aspect ratio and the stratification rather than by the time step alone. Numerical experiments confirm robustness in mesh resolution, aspect ratio and shift parameter for both the linear and the nonlinear problem, and a terrain-following flow-over-a-ridge test case demonstrates the solver on a deformed geometry.

Key words. Preconditioning, Boussinesq System, Compatible Finite Elements, Multigrid, Parameter Robustness, Implicit Time Stepping.

AMS subject classifications. 65N30 (finite element methods for BVPs), 65F08 (preconditioners for linear systems), 65N55 (multigrid methods), 76D07 (Stokes and related problems).

1. Introduction.

Numerical models of the ocean must integrate the governing equations over timescales far longer than the periods of the fastest waves the equations support. The incompressible Boussinesq system, which forms the dynamical core of many ocean models, admits internal gravity waves whose frequencies reach the buoyancy frequency N. In the small aspect ratio regime characteristic of ocean basins, where the horizontal extent exceeds the vertical extent by three or more orders of magnitude, these waves propagate quickly relative to the slow circulation that is the quantity of physical interest. An explicit time integration of the system is restricted by the Courant–Friedrichs–Lewy (CFL) condition tied to the gravity-wave speed, which forces time steps far smaller than accuracy alone would require. Implicit time integration removes this restriction, at the cost of solving a large linear system at every step. The efficiency of an implicit ocean model is therefore determined by the efficiency of the linear solver, and the design of a solver that remains effective and robust as the time step, the mesh resolution, and the aspect ratio vary is the problem addressed in this report. Operational ocean models have long avoided a fully implicit treatment of the fast waves by splitting the dynamics to isolate computationally stiff fast components. The classical strategy is barotropic–baroclinic (mode) splitting, in which the depth-averaged barotropic mode is stepped implicitly or sub-cycled at a

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small time step [11] while the residual baroclinic mode is integrated at the dynamical time step; this underpins models such as FESOM [10] and MITgcm [25]. An alternative is the split-explicit scheme used in MPAS-Ocean [21] and ROMS [30], in which the fast barotropic equations are sub-cycled within each slow step without a formal mode decomposition. Each of these approaches removes the gravity-wave CFL restriction from the fast component at comparatively low cost, since the barotropic or fast sub-problem has far fewer degrees of freedom than the full system, and they remain the standard approach in large-scale ocean circulation models. The splitting is, however, an approximation: it presumes a clean scale separation between the barotropic and baroclinic modes that is not always present, for instance in strongly stratified or strongly rotating regimes, or when internal gravity waves of frequency comparable to the time step of interest are themselves part of the dynamics being resolved. A fully implicit treatment of the system avoids this splitting error and remains accurate across the full range of wave frequencies the discretisation supports, at the cost that every time step now requires the solution of a saddle-point linear system coupling all three fields, rather than a sequence of smaller barotropic and baroclinic solves. The construction of a linear solver that is robust enough to make this fully implicit alternative competitive is the motivation for the remainder of the report.

For the purposes of this report we study the Boussinesq equations linearised about a state of rest, since the linearisation makes the theory tractable. The approach is, however, designed with the full nonlinear equations in view: there, the preconditioner is used for the iterative solution of the Jacobian system arising at each step of a Newton method, a setting exercised directly by the nonlinear experiments of Section 5. We work with a compatible finite element discretisation of the linearised nonhydrostatic Boussinesq equations. Compatible finite element methods preserve discrete analogues of the differential structure of the continuous equations, and have become a standard tool in geophysical fluid dynamics [6, 7]. The discretisation places velocity in an $H(\mathrm{div})$-conforming Raviart–Thomas space, pressure in a discontinuous space, and buoyancy in a tensor-product space chosen to avoid the spurious modes that arise from an incompatible pairing [27, 29]. The implicit time discretisation produces, at each step, a saddle-point linear system coupling velocity, buoyancy, and pressure. Two features of this system make it difficult to precondition. The velocity block is an $H(\mathrm{div})$ mass-type operator whose conditioning degrades on the anisotropic meshes that small aspect ratios demand, a difficulty already identified for the related elliptic problem on thin domains [23]. The saddle-point structure, with its zero pressure block, prevents a direct block factorisation and requires a Schur-complement treatment that the gravity-wave coupling renders sensitive to the time step.

The preconditioner developed here is built on operator preconditioning, in which a discrete operator is preconditioned by a norm-equivalent operator derived from the mapping properties of the continuous problem [22, 24]. We introduce a shifted perturbation of the saddle-point system that augments the pressure equation with a small multiple of the pressure mass matrix. The shift makes the pressure block invertible in isolation, which permits an exact Schur-complement elimination of the pressure and reduces the solve to a velocity-buoyancy block augmented by a grad-div term. The size of the shift controls the spectral distance between the shifted operator and the original one, and the central theoretical result of the report is a bound on this distance with explicit dependence on the time step, the buoyancy frequency, and the aspect ratio. The bound is obtained by reducing the shifted problem to a perturbed mixed problem and identifying the relevant inf-sup constant. A consequence of the bound is that the inf–sup constant is exactly independent of the aspect ratio throughout the

thin domain regime, so the preconditioner is robust precisely in the limit where standard approaches degrade. The bound also predicts the scaling of the shift parameter with the time step, and this prediction is recovered in the numerical experiments.

The velocity-buoyancy block that remains after the Schur-complement reduction is solved by a multigrid method with a patch-based line smoother. The smoother relaxes over overlapping patches of vertical columns of the extruded mesh, which captures the strong vertical coupling induced by the small aspect ratio. This patch structure differs from the monolithic patch smoothers applied directly to the full saddle-point system [16], in which each patch contains velocity and pressure degrees of freedom. Monolithic patches are robust but carry a high cost per application because of their size. The Schur-complement reduction used here yields smaller patches that contain only velocity-buoyancy degrees of freedom, and the efficient patch assembly available in Firedrake [20] makes the resulting solver practical. We report timings for the preconditioner application and examine the effect of a reordering of the patch subproblems on the cost of the local factorisations.

The contribution of the report is a shifted Schur-complement preconditioner for the implicit linearised Boussinesq equations that is robust in the time step, the mesh resolution, and the aspect ratio, together with an analysis that establishes this robustness and predicts the choice of the shift parameter. The work is a step towards an efficient implicit solver for ocean models, and is restricted to the linearised equations so that the parameter robustness of the linear solver can be studied in isolation from the outer nonlinear iteration. The linearised system is the inner solve in any Newton or Picard iteration for the full nonlinear problem, so its robust solution is a prerequisite for an efficient nonlinear solver.

Shifting an indefinite or singular operator to make it more amenable to a direct or iterative inner solve, and then using the shifted operator as a preconditioner for the original problem, is a strategy with substantial precedent outside the saddle-point setting considered here. The closest analogue is the shifted-Laplacian preconditioner for the Helmholtz equation, in which the indefinite Helmholtz operator is preconditioned by a complex-shifted variant that is definite and therefore amenable to multigrid [12, 13]; the shift parameter there plays a role directly analogous to δ here, trading proximity to the original operator against the conditioning of the auxiliary problem, and a substantial body of work has developed sharp estimates for this trade-off and for sweeping and domain-decomposition variants of the idea [17]. The shifted-Schur preconditioner developed below is the corresponding construction for the saddle-point structure of the Boussinesq system: the shift acts on the zero pressure block rather than on an indefinite term, but the underlying mechanism, a small perturbation that restores well-posedness of an auxiliary problem at a controllable spectral cost, is the same.

The rest of the report is organised as follows. Section 2 describes the governing equations, the compatible finite element discretisation, and the implicit time discretisation. Section 3 constructs the shifted Schur-complement preconditioner and the patch-based multigrid solver. Section 4 establishes the convergence bound and the asymptotic scaling of the shift parameter. Section 5 reports the numerical experiments, confirming parameter robustness and the predicted shift-parameter scaling. Section 6 concludes, and Section 7 sets out the directions for future work. Implementation details of the time stepper and the solver configuration are collected in the appendices.

2. Governing Equations and Discretisations.

2.1. Governing equations. We model large-scale oceanic flows using the incompressible Boussinesq system, linearised about a hydrostatic background state with linear stratification. The linearised system and its weak formulation below are posed on a general three-dimensional domain, since this is the setting in which the preconditioner of Section 3 and the theory of Section 4 are developed; the two-dimensional vertical slice in the x–z plane used for the compatible finite element discretisation and the numerical experiments of this report is introduced as a specialisation at the end of this section.

We begin from the three-dimensional incompressible Boussinesq system, consisting of the momentum equation,

$$\frac{\partial \vec{u}}{\partial t} + (\vec{u}\cdot\nabla)\vec{u} + 2\Omega\times\vec{u} + \nabla p - b\hat{k} = 0, \tag{2.1}$$

the incompressibility constraint,

$$\nabla\cdot\vec{u} = 0, \tag{2.2}$$

and the transport equation for buoyancy,

$$\frac{\partial b}{\partial t} + \vec{u}\cdot\nabla b = 0, \tag{2.3}$$

where $\vec{u} = (u, v, w)$ is the three-dimensional velocity vector, p is the pressure, b is the buoyancy, and $2\Omega = (0, 0, f)$ with f the constant Coriolis parameter. In general f varies with latitude, neither the discretisation nor the preconditioner relies on f being constant, since the Coriolis term enters only through the pointwise skew operator $\vec{u} \mapsto f\,\hat{k}\times\vec{u}$.

To linearise about a hydrostatic background state, we decompose each variable into a steady background state and a perturbation,

$$b = \bar{b} + b', \quad \vec{u} = \bar{\vec{u}} + \vec{u}', \quad p = \bar{p} + p', \tag{2.4}$$

with linear stratification $\bar{b}(z) = N^2(z)z$, where $N(z)$ is the buoyancy frequency that may depend on depth, and no background flow $\bar{\vec{u}} = (0, 0, 0)$ for simplicity. The background pressure $\bar{p}$ is determined up to a constant by the hydrostatic balance $\nabla\bar{p} - \bar{b}\hat{k} = 0$. Substituting the decomposition into (2.1)–(2.3) and discarding terms quadratic in the perturbations yields the linearised Boussinesq equations,

$$\frac{\partial \vec{u}'}{\partial t} + 2\Omega\times\vec{u}' + \nabla p' - b'\hat{k} = 0, \tag{2.5}$$

$$\frac{\partial b'}{\partial t} + N^2 w' = 0, \tag{2.6}$$

$$\nabla\cdot\vec{u}' = 0, \tag{2.7}$$

posed on $\Omega\times[0,T]$, where $\Omega = [0, L_x]\times[0, L_y]\times[0, L_z] \subset \mathbb{R}^3$ is a rectangular box domain with horizontal extents $L_x, L_y > 0$ and depth $L_z > 0$. The domain is periodic in the two horizontal directions x and y and bounded above and below by rigid lids. This specific choice of geometry and boundary conditions is an assumption made for the theoretical results of Section 4 rather than a limitation of the method. In particular, free-slip boundary conditions in the horizontal directions can be handled in the same way. The three-dimensional aspect ratios $\varepsilon_x = L_z/L_x$ and $\varepsilon_y = L_z/L_y$

measure the shallowness of the domain; in ocean basins, both are typically of order 10^{-3} or smaller. The boundary conditions are the no-flux condition $\vec{u}' \cdot \hat{n} = 0$ on the top and bottom boundaries $\partial\Omega_{\mathrm{top}} \cup \partial\Omega_{\mathrm{bot}}$. We have used the linear stratification $\bar{b}(z) = N^2 z$ to write $w'\,\partial_z \bar{b} = N^2 w'$ in (2.6).

To prepare for the compatible finite element discretisation of Section 2.2 and the preconditioner and theory of Sections 3 and 4, we cast the linearised system (2.5)–(2.7) in weak form, on the general three-dimensional domain Ω introduced above. Multiplying the equation by test functions, integrating over Ω, and integrating the pressure-gradient term by parts, we obtain the following weak form: find $\vec{u}'(t) \in \mathring{H}(\mathrm{div})$, $b'(t) \in L^2$, $p'(t) \in \mathring{L}^2(\Omega)$ such that for all $(\vec{w}, q, \phi) \in \mathring{H}(\mathrm{div}) \times \mathbb{W}^\theta \times \mathring{L}^2(\Omega)$,

$$\left\langle \vec{w},\, \frac{\partial \vec{u}'}{\partial t} \right\rangle + \langle \vec{w},\, 2\Omega \times \vec{u}' \rangle - \langle \nabla \cdot \vec{w},\, p' \rangle - \left\langle \vec{w},\, b' \hat{k} \right\rangle = 0, \tag{2.8a}$$

$$\left\langle q,\, \frac{\partial b'}{\partial t} \right\rangle + \left\langle q,\, N^2 \vec{u}' \cdot \hat{k} \right\rangle = 0, \tag{2.8b}$$

$$\langle \phi,\, \nabla \cdot \vec{u}' \rangle = 0, \tag{2.8c}$$

where $\langle \cdot, \cdot \rangle$ denotes the $L^2(\Omega)$ inner product, applied componentwise where appropriate, and

$$\mathring{H}(\mathrm{div}) = \{ \vec{w} \in H(\mathrm{div}; \Omega) : \vec{w} \cdot \hat{n} = 0 \text{ on } \partial\Omega_{\mathrm{top}} \cup \partial\Omega_{\mathrm{bot}} \} \tag{2.9}$$

encodes the no-flux boundary condition as an essential condition.

We define the mean-zero pressure space

$$\mathring{L}^2(\Omega) := \left\{ p \in L^2(\Omega) : \textstyle\int_\Omega p \,\mathrm{d}x = 0 \right\}. \tag{2.10}$$

The restriction $p'(t) \in \mathring{L}^2(\Omega)$ is the correct function space for the pressure: since $\nabla \cdot \vec{w}$ has zero mean for every $\vec{w} \in \mathring{H}(\mathrm{div})$ on a domain with no inflow or outflow boundary, the divergence equation (2.8c) only constrains p' up to a global additive constant. Imposing $p' \in \mathring{L}^2(\Omega)$ removes this redundancy and renders the variational problem (2.8)–(2.8c) uniquely solvable. This single global mean-zero constraint has no localised basis representation and is the source of the saddle-point structure discussed in Section 3 below; we return to how it is affected by the slice reduction that follows, and by the passage to the discrete setting, once the block structure of the discrete system has been introduced.

For the numerical experiments of this report we specialise the three-dimensional system above to a two-dimensional vertical slice in the x–z plane as a simplified model problem. In the vertical slice model, we restrict the spatial dependence of the perturbed variables to the x–z plane while retaining all three Cartesian components of the velocity field. With $\partial_y \equiv 0$ on every perturbed variable, the gradient and divergence operators in (2.5)–(2.7) act only in x and z,

$$\nabla \phi = \left(\frac{\partial \phi}{\partial x},\, 0,\, \frac{\partial \phi}{\partial z} \right), \qquad \nabla \cdot \vec{u}' = \frac{\partial u'}{\partial x} + \frac{\partial w'}{\partial z}, \tag{2.11}$$

and, reusing the symbol Ω for the remainder of the report to denote this two-dimensional slice domain, we take $\Omega = [0, L_x] \times [0, L_z] \subset \mathbb{R}^2$ with aspect ratio

$$\varepsilon = \frac{L_z}{L_x}, \tag{2.12}$$

which coincides with ε_x from the three-dimensional domain above. In the oceanic regime motivating this work, ε is typically of order 10^{-3} or smaller. The slice form of the linearised system (2.5)–(2.7) reads

$$\frac{\partial u'}{\partial t} - fv' + \frac{\partial p'}{\partial x} = 0, \tag{2.13a}$$

$$\frac{\partial v'}{\partial t} + fu' = 0, \tag{2.13b}$$

$$\frac{\partial w'}{\partial t} + \frac{\partial p'}{\partial z} - b' = 0, \tag{2.13c}$$

$$\frac{\partial b'}{\partial t} + N^2 w' = 0, \tag{2.13d}$$

$$\frac{\partial u'}{\partial x} + \frac{\partial w'}{\partial z} = 0, \tag{2.13e}$$

on $\Omega \times [0, T]$, with the boundary and initial conditions inherited from the three-dimensional system above. The weak form (2.8)–(2.8c) specialises directly to the slice: $H(\mathrm{div}; \Omega)$, $\mathbb{W}^\theta$, and $L^2(\Omega)$ are simply taken on the two-dimensional slice domain.

The system (2.13) exposes the two coupled wave processes that drive the dynamics and motivate the implicit time stepping used in this report. Equations (2.13c) and (2.13d), together with the divergence constraint (2.13e), support internal gravity waves with frequency bounded above by the buoyancy frequency N, while (2.13a)–(2.13b) carry inertial oscillations at the Coriolis frequency f. In oceanic applications, the timescales of interest are typically those of the slow, large-scale circulation, while the fastest internal gravity waves are several orders of magnitude faster. Resolving these waves explicitly imposes a severe CFL restriction that is decoupled from any accuracy requirement on the dynamics one actually wishes to track. Moreover, ocean basins exhibit aspect ratios ε that are typically much smaller than their atmospheric counterparts, which sharpens the anisotropy of the discrete operator and makes solver robustness in ε a practical requirement.

2.2. Finite element discretisation. We discretise the weak form (2.8) in space using a compatible finite element triple on an extruded quadrilateral mesh, following the framework developed in [7]. The temporal discretisation, described in Section 2.3, uses a fully implicit midpoint rule.

In the three-dimensional setting, the domain $\Omega = [0, L_x] \times [0, L_y] \times [0, L_z]$ is discretised by an extruded mesh: a two-dimensional structured horizontal base mesh is extruded vertically into N_z layers, producing a hexahedral mesh whose cells inherit the anisotropy of the small aspect ratios $\varepsilon_x, \varepsilon_y$. The tensor-product structure of the extrusion carries over directly to the finite element spaces, facilitating compatible discretisations in three dimensions. For the two-dimensional vertical-slice experiments of this report, the slice domain $\Omega = [0, L_x] \times [0, L_z]$ is discretised by the corresponding lower-dimensional construction: a one-dimensional base mesh of N_x intervals in the horizontal direction is extruded vertically into N_z layers, producing a structured quadrilateral mesh on Ω with cell widths $\Delta x = L_x/N_x$ and $\Delta z = L_z/N_z$. The extruded construction produces the strongly anisotropic cells associated with small aspect ratios $\varepsilon = L_z/L_x$ without the mesh-generation difficulties of unstructured meshes in thin domains. It also brings two practical advantages that the solver of Section 3 relies on. The tensor-product finite element spaces the extrusion induces are generated symbolically from their horizontal and vertical factors [26], so that the triple below is assembled without hand-coding the product elements; and the degrees

of freedom are numbered so as to exploit the fixed column structure of the mesh [3], which makes the data belonging to a vertical column contiguous. The latter is what allows the column-patch smoother of Section 3.4 to be assembled and applied as a local operation.

We adopt the compatible finite element triple of [27, 29], as surveyed in [7]. For polynomial degree $k \geq 1$, the discrete spaces for velocity, buoyancy and pressure on the extruded quadrilateral mesh are

$$H_h(\mathrm{div}) = \mathrm{RT}_k, \qquad \mathbb{W}_h^\theta = \mathrm{DG}_{k-1} \otimes \mathrm{CG}_k, \qquad L_h^2 = \mathrm{DG}_{k-1}, \tag{2.14}$$

where $H_h(\mathrm{div}) \subset H(\mathrm{div};\Omega)$ is the Raviart–Thomas space, $L_h^2 \subset L^2(\Omega)$ is the fully discontinuous pressure space with mean-zero subspace $\mathring{L}_h^2 = \{p_h \in L_h^2 : \int_\Omega p_h \,\mathrm{d}x = 0\}$, and $\mathbb{W}_h^\theta$ is the tensor-product buoyancy space, discontinuous in the horizontal direction and continuous in the vertical. The vertical continuity of $\mathbb{W}_h^\theta$ aligns the buoyancy degrees of freedom with the vertical component of $H_h(\mathrm{div})$ on horizontal facets, eliminating the spurious modes that would arise from a fully discontinuous buoyancy space; we refer to [27] and to [7, Section 5] for a detailed discussion.

The no-flux boundary condition is imposed as an essential condition on $H_h(\mathrm{div})$. We write

$$\mathring{H}_h(\mathrm{div}) = \{\vec{w} \in H_h(\mathrm{div}) : \vec{w} \cdot \hat{n} = 0 \text{ on } \partial\Omega_{\mathrm{top}} \cup \partial\Omega_{\mathrm{bot}}\}\,. \tag{2.15}$$

Periodicity in the horizontal direction is built into the base mesh and is inherited by all three spaces. In the numerical experiments of Section 5 we take $k = 2$, giving the second-order $\mathrm{RT}_2 \otimes \mathbb{W}_h^\theta \otimes \mathrm{DG}_1$ triple.

The semi-discrete problem reads: find $\vec{u}_h'(t) \in \mathring{H}_h(\mathrm{div})$, $b_h'(t) \in \mathbb{W}_h^\theta$, $p_h'(t) \in \mathring{L}_h^2$ such that for all $(\vec{w}, q, \phi) \in \mathring{H}_h(\mathrm{div}) \times \mathbb{W}_h^\theta \times \mathring{L}_h^2$,

$$\left\langle \vec{w},\, \frac{\partial \vec{u}_h'}{\partial t} \right\rangle + \langle \vec{w},\, 2\Omega \times \vec{u}_h' \rangle - \langle \nabla \cdot \vec{w},\, p_h' \rangle - \left\langle \vec{w},\, b_h' \hat{k} \right\rangle = 0, \tag{2.16a}$$

$$\left\langle q,\, \frac{\partial b_h'}{\partial t} \right\rangle + \left\langle q,\, N^2 \vec{u}_h' \cdot \hat{k} \right\rangle = 0, \tag{2.16b}$$

$$\langle \phi,\, \nabla \cdot \vec{u}_h' \rangle = 0. \tag{2.16c}$$

The compatibility of the triple (2.14) ensures that the divergence operator maps $H_h(\mathrm{div})$ surjectively onto L_h^2, so the discrete inf–sup condition holds and the divergence constraint (2.16c) is discretised consistently. This compatibility is the structural property exploited by the preconditioner constructed in Section 3.

2.3. Time discretisation. The semi-discrete system (2.16) is integrated in time using a fully implicit midpoint rule. Writing $\bar{(\cdot)} := \tfrac{1}{2}\big((\cdot)^{n+1} + (\cdot)^n\big)$ for the midpoint average, the fully discrete update from t_n to $t_{n+1} = t_n + \Delta t$ reads: find $(\vec{u}_h'^{\,n+1}, b_h'^{\,n+1}, p_h'^{\,n+1}) \in \mathring{H}_h(\mathrm{div}) \times \mathbb{W}_h^\theta \times \mathring{L}_h^2$ such that

$$\begin{aligned} &\left\langle \vec{w},\, \vec{u}_h'^{\,n+1} - \vec{u}_h'^{\,n} \right\rangle + \Delta t \left\langle \vec{w},\, 2\Omega \times \bar{\vec{u}}_h' \right\rangle - \Delta t \left\langle \nabla \cdot \vec{w},\, p_h'^{\,n+1} \right\rangle \\ &\qquad - \Delta t \left\langle \vec{w},\, \bar{b}_h' \hat{k} \right\rangle = 0, \end{aligned} \tag{2.17a}$$

$$\left\langle q,\, b_h'^{\,n+1} - b_h'^{\,n} \right\rangle + \Delta t \left\langle q,\, N^2 \bar{\vec{u}}_h' \cdot \hat{k} \right\rangle = 0, \tag{2.17b}$$

$$\left\langle \phi,\, \nabla \cdot \vec{u}_h'^{\,n+1} \right\rangle = 0, \tag{2.17c}$$

for all $(\vec{w}, q, \phi) \in \mathring{H}_h(\mathrm{div}) \times \mathbb{W}_h^\theta \times \mathring{L}_h^2$. The system (2.17) is the linear saddle-point problem whose block structure is analysed and preconditioned in Sections 3 and 4.

3. Preconditioning strategy. The fully discrete system (2.17) is a linear saddle-point problem at each time step. In this section we construct the preconditioner that is the main contribution of the report. The construction has three structural ingredients: a shifted perturbation of the divergence equation that makes the pressure block invertible in isolation, a Schur complement elimination of the pressure variable that yields a transport-type block on the velocity and buoyancy fields, and a patch-based multigrid smoother that delivers ε-robust inner solves on the resulting block. We treat each in turn.

3.1. Block structure of the fully discrete system. Throughout this section, matrix notation is used only to display the block structure of the solver: matrices, written in plain letters (A_0, A_δ, S, ...), act on vectors of basis coefficients. The analysis of Section 4 is carried out entirely at the variational level, where the corresponding operators are written calligraphically ($\mathcal{A}_0$, $\mathcal{A}_\delta$). Writing $\vec{x}$ for the coefficient vector of the unknown $(\vec{u}_h'^{\,n+1}, b_h'^{\,n+1}, p_h'^{\,n+1})$ and $\vec{f}$ for the load vector assembled from the data at time level t_n, the system (2.17) can be cast as the linear system

$$A_0 \vec{x} = \vec{f}, \tag{3.1}$$

where A_0 is the 3×3 block matrix

$$A_0 = \begin{pmatrix} M_u + \frac{\Delta t}{2} C & -\frac{\Delta t}{2} J & -\Delta t\, G \\ \frac{\Delta t}{2} N^2 J^\top & M_b & 0 \\ G^\top & 0 & 0 \end{pmatrix}, \tag{3.2}$$

with the velocity mass matrix M_u assembled from $\langle \vec{w}, \vec{u}_h' \rangle$, the buoyancy mass matrix M_b from $\langle q, b_h' \rangle$, the Coriolis matrix C from $\langle \vec{w}, 2\Omega \times \vec{u}_h' \rangle$, the vertical-coupling matrix J from $\langle \vec{w}, b_h' \hat{k} \rangle = \langle \vec{w} \cdot \hat{k}, b_h' \rangle$, and the divergence matrix $G^\top$ from $\langle \phi, \nabla \cdot \vec{u}_h' \rangle$, with $-G$ assembled from $-\langle \nabla \cdot \vec{w}, p_h' \rangle$.

The system (3.2) exhibits two distinct sources of difficulty that compound each other. First, the $(3,3)$ block is zero, so A_0 is a saddle-point matrix and the pressure cannot be eliminated by a naive block factorisation. The zero block is the discrete trace of the global mean-zero constraint on $p_h' \in \mathring{L}_h^2$ identified at the continuous level after the weak formulation (2.8) above. Because this is a single global constraint, it has no finite element basis representation; in our implementation the constant nullspace is projected out globally at the Krylov level. By itself this would be a mild inconvenience. Second, the genuine difficulty is what happens as the aspect ratio $\varepsilon \to 0$: the horizontal derivative ∂_x becomes subdominant to ∂_z, so in the thin-domain limit the divergence constraint decouples column by column. Each vertical column then carries its own depth-independent pressure mode, and the single global nullspace of the well-posed problem grows to dimension comparable to the number of horizontal degrees of freedom. It is this growth of the effective nullspace under anisotropy, not the bare presence of a pressure nullspace, that makes A_0 difficult to precondition robustly in ε. The $(1,1)$ block compounds this: dominated by the $H(\mathrm{div})$ mass operator M_u, it is poorly conditioned on extruded meshes with small aspect ratio [7, 23], since mesh anisotropy translates directly into anisotropy of the $H(\mathrm{div})$ inner product. Any preconditioner for A_0 must address both difficulties.

3.2. The shifted system. We resolve the saddle-point structure by introducing a shifted preconditioner that modifies the divergence equation of the fully discrete system (2.17) by augmenting it with a pressure mass term. Specifically, the shifted variational problem reads: find $(\vec{u}_h'^{\,n+1}, b_h'^{\,n+1}, p_h'^{\,n+1}) \in \mathring{H}_h(\mathrm{div}) \times \mathbb{W}_h^\theta \times \mathring{L}_h^2$ such that

for all $(\vec{w}, q, \phi) \in \mathring{H}_h(\mathrm{div}) \times \mathbb{W}_h^\theta \times \mathring{L}_h^2$ the momentum and buoyancy equations are as in (2.17), while the divergence constraint is replaced by

$$\langle \phi,\, \nabla \cdot \vec{u}_h'^{\,n+1} \rangle + \delta \, \langle \phi,\, p_h'^{\,n+1} \rangle = 0, \tag{3.3a}$$

where $\delta > 0$ is the shift parameter. The system defined by (2.17a)–(2.17b) and (3.3a) is used as a preconditioner for the original system $A_0 \vec{x} = \vec{f}$; we apply a Krylov method to $A_0 \vec{x} = \vec{f}$ with the solution operator of the shifted system as the preconditioner. In block-matrix notation, with the same ordering of the unknowns, the matrix of the shifted system is

$$A_\delta = \begin{pmatrix} M_u + \frac{\Delta t}{2} C & -\frac{\Delta t}{2} J & -\Delta t\, G \\ \frac{\Delta t}{2} N^2 J^\top & M_b & 0 \\ G^\top & 0 & \delta M_p \end{pmatrix}, \tag{3.4}$$

which differs from A_0 only by the augmentation of the $(3,3)$ block with the pressure mass matrix δM_p, corresponding to the additional term $\delta \langle \phi, p_h' \rangle$ in the divergence equation. The shift parameter $\delta > 0$ is a small constant whose choice is discussed below. The shifted matrix A_δ is used as a preconditioner for A_0.

The motivation for the shifted preconditioner is threefold. First, the $(3,3)$ block of A_δ is the pressure mass matrix δM_p, which is block-diagonal, the pressure space being discontinuous, and therefore trivially invertible. This enables an exact Schur complement elimination of the pressure variable, as carried out in Section 3.3 below. Second, the perturbation $A_\delta - A_0$ acts only on the pressure variable and has magnitude $O(\delta)$, so A_δ remains spectrally close to A_0 for small δ. The quantitative form of this statement, $\|\mathcal{A}_\delta^{-1}\mathcal{A}_0 - I\| \le l(\delta, \Delta t, N^2)$ for some function l under an appropriate choice of norm, with $\mathcal{A}_0$ and $\mathcal{A}_\delta$ the operators induced by the unshifted and shifted variational problems, is established in Section 4; for the purposes of constructing the preconditioner here, it suffices to know that the bound is small and δ-controlled. Third, the shift directly removes the growing pressure nullspace described above: δM_p is strictly positive definite on all of L_h^2, including the constant mode and, in the thin-domain limit, the column-wise depth-independent modes, so A_δ has no pressure nullspace for any $\delta > 0$ and any aspect ratio, independently of the global nullspace treatment required for A_0 itself.

3.3. Schur complement elimination of pressure. The elimination of pressure proceeds directly in the variational setting. The shifted divergence constraint (3.3a) reads: for all $\phi \in \mathring{L}_h^2$,

$$\langle \phi,\, \nabla \cdot \vec{u}_h' \rangle = -\delta \langle \phi,\, p_h' \rangle. \tag{3.5}$$

Since the compatible triple (2.14) satisfies $\nabla \cdot \mathring{H}_h(\mathrm{div}) \subseteq \mathring{L}_h^2$, the function $\nabla \cdot \vec{u}_h' + \delta\, p_h'$ lies in $\mathring{L}_h^2$; equation (3.5) states that it is L^2-orthogonal to all of $\mathring{L}_h^2$, so it vanishes identically, determining p_h' from $\vec{u}_h'$ as a finite element function:

$$p_h' = -\frac{1}{\delta} \nabla \cdot \vec{u}_h'. \tag{3.6}$$

No mass-matrix inverse appears at this level: the L^2 projection of $\nabla \cdot \vec{u}_h'$ onto the pressure space is $\nabla \cdot \vec{u}_h'$ itself. Substituting (3.6) into the pressure-gradient term of the momentum equation replaces the saddle-point coupling by a grad-div term acting on the velocity alone:

$$-\langle \nabla \cdot \vec{w},\, p_h' \rangle \;\mapsto\; +\frac{1}{\delta} \langle \nabla \cdot \vec{w},\, \nabla \cdot \vec{u}_h' \rangle. \tag{3.7}$$

The pressure variable is now expressed entirely through $\vec{u}_h'^{n+1}$ via (3.6); the two-field system for $(\vec{u}_h'^{n+1}, b_h'^{n+1}) \in \mathring{H}_h(\mathrm{div}) \times \mathbb{W}_h^\theta$ reads: find $(\vec{u}_h'^{n+1}, b_h'^{n+1})$ such that for all $(\vec{w}, q) \in \mathring{H}_h(\mathrm{div}) \times \mathbb{W}_h^\theta$,

(3.8a)
$$\left\langle \vec{w},\, \vec{u}_h'^{n+1} - \vec{u}_h'^{n} \right\rangle + \Delta t \left\langle \vec{w},\, 2\Omega \times \bar{\vec{u}}_h' \right\rangle + \frac{\Delta t}{\delta} \left\langle \nabla \cdot \vec{w},\, \nabla \cdot \vec{u}_h'^{n+1} \right\rangle - \Delta t \left\langle \vec{w},\, \bar{b}_h' \hat{k} \right\rangle = 0,$$

(3.8b)
$$\left\langle q,\, b_h'^{n+1} - b_h'^{n} \right\rangle + \Delta t \left\langle q,\, N^2 \bar{\vec{u}}_h' \cdot \hat{k} \right\rangle = 0,$$

where the grad-div term $+\frac{\Delta t}{\delta}\langle \nabla\cdot\vec{w}, \nabla\cdot\vec{u}_h'^{n+1}\rangle$ replaces the pressure-gradient coupling of (2.17a). The system (3.8) carries no pressure variable and therefore no nullspace. In block-matrix notation this elimination is the Schur complement of A_δ with respect to its pressure block; that block form, which is used to state the preconditioner and to assemble it in practice, is introduced next.

To exploit the structure of A_δ we collapse the velocity-buoyancy variables into a single transport block and treat pressure separately. Writing $\vec{y} = (\vec{u}_h', b_h') \in \mathring{H}_h(\mathrm{div}) \times \mathbb{W}_h^\theta$, the shifted system takes the 2×2 block form

$$A_\delta = \begin{pmatrix} D & B \\ E & \delta M_p \end{pmatrix}, \tag{3.9}$$

where D is the transport block acting on $\vec{y}$,

$$D = \begin{pmatrix} M_u + \frac{\Delta t}{2} C & -\frac{\Delta t}{2} J \\ \frac{\Delta t}{2} N^2 J^\top & M_b \end{pmatrix}, \tag{3.10}$$

$B = (-\Delta t\, G,\, 0)^\top$ is the velocity-pressure coupling, and $E = (G^\top,\, 0) = -\Delta t^{-1} B^\top$ is its counterpart on the divergence side.

Because the shift has made the pressure block invertible, block elimination in the sense of [19, Section 3.2] may be applied to it, and the Schur complement of A_δ with respect to the pressure block is, formally,

$$S = D - B(\delta M_p)^{-1} E = D - \delta^{-1}\, B M_p^{-1} E. \tag{3.11}$$

Because M_p is block-diagonal, with one dense block per element, the inverse M_p^{-1} is computed exactly by a single direct solve per element. The action of $-\delta^{-1} B M_p^{-1} E$ on the velocity-buoyancy variable $\vec{y}$ is therefore assemblable in closed form. With $B = (-\Delta t\, G, 0)^\top$ and $E = (G^\top, 0)$, the product $-\delta^{-1} B M_p^{-1} E$ contributes only to the velocity diagonal and corresponds, in the weak form, to the replacement

$$-\left\langle \nabla\cdot\vec{w},\, p_h' \right\rangle \;\longrightarrow\; +\frac{1}{\delta}\left\langle \nabla\cdot\vec{w},\, \nabla\cdot\vec{u}_h' \right\rangle, \tag{3.12}$$

with no contribution to the buoyancy block. The exact Schur complement (3.11) is therefore

$$S = \begin{pmatrix} M_u + \frac{\Delta t}{2} C + \frac{\Delta t}{\delta} G M_p^{-1} G^\top & -\frac{\Delta t}{2} J \\ \frac{\Delta t}{2} N^2 J^\top & M_b \end{pmatrix}, \tag{3.13}$$

where the grad-div correction $+\frac{\Delta t}{\delta} G M_p^{-1} G^\top$ is the assembled counterpart of substituting (3.6) into the pressure-gradient term $-\Delta t\langle \nabla\cdot\vec{w}, p_h'\rangle$ of the momentum equation.

At the matrix level, the inverse of the shifted matrix factorises as

$$A_\delta^{-1} = \begin{pmatrix} I & D^{-1}B \\ 0 & I \end{pmatrix}^{-1} \begin{pmatrix} D^{-1} & 0 \\ 0 & S^{-1} \end{pmatrix} \begin{pmatrix} I & 0 \\ ED^{-1} & I \end{pmatrix}^{-1}, \tag{3.14}$$

so applying A_δ^{-1} requires only solves with D (or equivalently S on the velocity-buoyancy block, since the pressure mass solve is exact) and triangular block applications.

Unlike A_0, the system (3.8) involves no pressure variable and therefore carries no nullspace. We establish well-posedness via the Lax–Milgram theorem on the product space $Z_h = \mathring{H}_h(\mathrm{div}) \times \mathbb{W}_h^\theta$ equipped with the parameter-weighted norm

$$\|(\vec{v}, r)\|_Z^2 := \|\vec{v}\|_{L^2}^2 + \|\nabla \cdot \vec{v}\|_{L^2}^2 + N^{-2}\|r\|_{L^2}^2. \tag{3.15}$$

The N^{-2} weighting of the buoyancy component is the standard non-dimensionalisation for stratified flows; it exposes the skew-symmetry of the velocity–buoyancy coupling. This norm and the solvability result below serve only to establish well-posedness of the inner velocity–buoyancy solve, and are independent of the analysis of Section 4, which is carried out in the stratification-weighted norm (4.15) on the velocity–pressure space.

Writing $\vec{u} := \vec{u}_h'^{\,n+1}$, $b := b_h'^{\,n+1}$, $\vec{u}^n := \vec{u}_h'^{\,n}$, $b^n := b_h'^{\,n}$ for brevity, and dividing the buoyancy equation (3.8b) by N^2, which is what renders the velocity–buoyancy coupling skew-symmetric in the proof below, the system (3.8) takes the abstract form: find $(\vec{u}, b) \in Z_h$ such that

$$\mathcal{A}\big((\vec{u}, b),\, (\vec{w}, q)\big) = \mathcal{F}(\vec{w}, q) \qquad \forall\, (\vec{w}, q) \in Z_h, \tag{3.16}$$

where

$$\mathcal{A}\big((\vec{u}, b),\, (\vec{w}, q)\big) := a_u(\vec{u}, \vec{w}) + a_b(b, q) + c(\vec{u}, q) + d(b, \vec{w}), \tag{3.17}$$

$$a_u(\vec{u}, \vec{w}) := \langle \vec{u}, \vec{w}\rangle + \tfrac{\Delta t}{2}\langle 2\Omega \times \vec{u},\, \vec{w}\rangle + \tfrac{\Delta t}{\delta}\langle \nabla \cdot \vec{u},\, \nabla \cdot \vec{w}\rangle, \tag{3.18}$$

$$a_b(b, q) := \tfrac{1}{N^2}\langle b, q\rangle, \quad c(\vec{u}, q) := \tfrac{\Delta t}{2}\langle \hat{k} \cdot \vec{u},\, q\rangle, \quad d(b, \vec{w}) := -\tfrac{\Delta t}{2}\langle b\hat{k},\, \vec{w}\rangle, \tag{3.19}$$

and $\mathcal{F} \in Z_h'$ collects the data from time level t_n:

$$\mathcal{F}(\vec{w}, q) := \langle \vec{u}^n, \vec{w}\rangle - \tfrac{\Delta t}{2}\langle 2\Omega \times \vec{u}^n,\, \vec{w}\rangle + \tfrac{\Delta t}{2}\langle b^n \hat{k},\, \vec{w}\rangle + \tfrac{1}{N^2}\langle b^n, q\rangle - \tfrac{\Delta t}{2}\langle \hat{k} \cdot \vec{u}^n,\, q\rangle. \tag{3.20}$$

PROPOSITION 3.1 (Solvability of the Schur-reduced system). *Let $N^2 > 0$, $\Delta t > 0$ and $\delta > 0$. For any $\vec{u}^n \in \mathring{H}_h(\mathrm{div})$ and $b^n \in \mathbb{W}_h^\theta$, the system (3.8) has a unique solution $(\vec{u}, b) \in Z_h$ satisfying the stability estimate*

$$\|(\vec{u}, b)\|_Z \le \frac{1}{\alpha_0}\|\mathcal{F}\|_{Z'}, \qquad \alpha_0 := \min\left(1,\, \frac{\Delta t}{\delta}\right). \tag{3.21}$$

Proof. We verify the hypotheses of the Lax–Milgram theorem.

Continuity. Continuity follows by bounding the four contributions to $\mathcal{A}$ separately. The symmetric positive terms of a_u and a_b are precisely those defining $\|\cdot\|_Z$, so the Cauchy-Schwarz inequality bounds them by $\|(\vec{u}, b)\|_Z \|(\vec{w}, q)\|_Z$. Writing $f := |2\vec{\Omega}|$, the pointwise bound $|2\vec{\Omega} \times \vec{u}| \le f|\vec{u}|$ gives

$$\frac{\Delta t}{2}\big|\langle 2\vec{\Omega} \times \vec{u},\, \vec{w}\rangle\big| \le \frac{\Delta t f}{2}\, \|\vec{u}\|\, \|\vec{w}\|, \tag{3.22}$$

and $|\hat{k}| = 1$ gives $|c(\vec{u},q)| + |d(b,\vec{w})| \le \frac{\Delta t}{2}(\|\vec{u}\|\|q\| + \|b\|\|\vec{w}\|)$. Since $\|\vec{u}\| \le \|(\vec{u},b)\|_Z$ and $N^{-1}\|q\| \le \|(\vec{w},q)\|_Z$, with the analogous bounds for the remaining factors, these terms are controlled by $\frac{\Delta t f}{2}$ and $\Delta t N$ multiples of $\|(\vec{u},b)\|_Z\|(\vec{w},q)\|_Z$ respectively. Hence

$$\big|\mathcal{A}\big((\vec{u},b),\,(\vec{w},q)\big)\big| \le \Big(1 + \frac{\Delta t f}{2} + \Delta t N\Big)\|(\vec{u},b)\|_Z\,\|(\vec{w},q)\|_Z, \tag{3.23}$$

so $\mathcal{A}$ is continuous with a constant independent of δ and of the mesh.

Coercivity. Setting $(\vec{w},q) = (\vec{u},b)$: the Coriolis term vanishes since $2\Omega \times \vec{u}$ is pointwise orthogonal to $\vec{u}$, giving

$$a_u(\vec{u},\vec{u}) = \|\vec{u}\|_{L^2}^2 + \tfrac{\Delta t}{\delta}\|\nabla\cdot\vec{u}\|_{L^2}^2. \tag{3.24}$$

Both terms are non-negative; the grad-div term is non-negative because the Schur elimination produces the positive coefficient $+\Delta t/\delta$. For the coupling blocks,

$$c(\vec{u},b) + d(b,\vec{u}) = \tfrac{\Delta t}{2}\langle\hat{k}\cdot\vec{u},\,b\rangle - \tfrac{\Delta t}{2}\langle b\hat{k},\,\vec{u}\rangle = 0, \tag{3.25}$$

so the velocity–buoyancy coupling is skew-symmetric and drops out. Summing all four contributions,

$$\mathcal{A}\big((\vec{u},b),(\vec{u},b)\big) = \|\vec{u}\|_{L^2}^2 + \tfrac{\Delta t}{\delta}\|\nabla\cdot\vec{u}\|_{L^2}^2 + \tfrac{1}{N^2}\|b\|_{L^2}^2 \ge \alpha_0\,\|(\vec{u},b)\|_Z^2, \tag{3.26}$$

with $\alpha_0 = \min(1,\Delta t/\delta) > 0$. By the Lax–Milgram theorem there exists a unique solution, and testing with $(\vec{w},q) = (\vec{u},b)$ gives $\alpha_0\|(\vec{u},b)\|_Z^2 \le \mathcal{F}(\vec{u},b) \le \|\mathcal{F}\|_{Z'}\|(\vec{u},b)\|_Z$, from which (3.21) follows. ∎

3.4. Inner solvers and the patch-smoothed multigrid. On meshes with small aspect ratio ($\Delta x \gg \Delta z$), the discrete operator of the Schur complement on the velocity-buoyancy block S exhibits strong coupling in the vertical direction and weak coupling in the horizontal. Point-wise smoothers fail because they cannot resolve the strong vertical coupling, leading to poor smoothing of vertically-oscillating error modes. We instead use a vertex-star patch smoother on the extruded mesh: for each base-mesh vertex it gathers the degrees of freedom of the vertical columns sharing that vertex into one overlapping patch. Solving each patch exactly inverts the strong vertical coupling, effectively acting as a vertical line smoother, while the remaining weak horizontal coupling is handled by the multigrid hierarchy through horizontal coarsening. This combination of line smoothing and multigrid is robust with respect to the mesh aspect ratio, as we now make precise.

Let $X_h = \mathring{H}_h(\mathrm{div}) \times \mathbb{W}_h^\theta$ denote the velocity-buoyancy space on which S acts, and let $\{v_i\}_{i=1}^{N_x}$ be the vertices of the one-dimensional base mesh. For each base-mesh vertex v_i, let $X_i \subset X_h$ be the subspace spanned by the degrees of freedom of X_h supported on the patch ω_i: the vertical columns of cells sharing v_i (two columns for an interior vertex of the one-dimensional base mesh, one on each side of v_i). This is an overlapping decomposition $X_h = \sum_i R_i^\top X_i$, where $R_i : X_h \to X_i$ is the restriction to the patch. Writing $s(\cdot,\cdot)$ for the bilinear form associated with S, each patch carries a local operator $A_i : X_i \to X_i^*$ defined by $\langle A_i x_i, y_i\rangle = s(R_i^\top x_i, R_i^\top y_i)$ for $x_i, y_i \in X_i$, and the additive Schwarz preconditioner

$$B_{\mathrm{ASM}} = \sum_{i=1}^{N_x} R_i^\top A_i^{-1} R_i \tag{3.27}$$

approximates S^{-1} on X_h. The patch smoother is the corresponding damped Richardson iteration: given the current iterate x and right-hand side f, one relaxation sweep on each multigrid level updates

$$x \leftarrow x + \theta\, B_{\mathrm{ASM}}(f - Sx), \tag{3.28}$$

with damping factor θ. This construction, gathering all degrees of freedom touching a fixed topological entity of the mesh (here, a base-mesh vertex) into a patch and extruding the patch through the column, is precisely the *vertex-star* patch in the topological taxonomy of the PCPatch framework of Farrell, Knepley, Mitchell, and Wechsung [15], of which our column-patch smoother is a direct instance specialised to the extruded mesh.

Two properties of this smoother matter, and they rest on different arguments. The first is solvability of each local problem A_i. The restriction of (3.13) to a patch is a velocity–buoyancy problem of exactly the form treated in Proposition 3.1, posed on the patch rather than on the whole domain; it inherits the coercivity and continuity constants established there, so the Lax–Milgram theorem gives a unique local solution and each A_i is invertible.

The second is the parameter-robustness of the multigrid V-cycle, and here the choice of a vertex-star patch is what secures it for every degree k; for $k = 2$ a single-column patch is also found to work in practice (Section 5.7). The grad–div term in (3.13) carries a coefficient that grows as the shift $\delta \to 0$, so the velocity block approaches the grad–div-dominated regime in which robustness requires that the kernel of the grad–div operator—the divergence-free fields—admit a basis each of whose members is supported on a single patch [1, 15]. On the base mesh that kernel is spanned by the $\nabla^\perp$ of the continuous CG_k basis, and each such function is supported on the star of a base-mesh vertex; this is precisely why the patch is built as a vertex star. This argument fixes the horizontal footprint of the patch but not its vertical extent: a complete robustness proof through the columns, in the anisotropic $\varepsilon \to 0$ limit, is not yet available, and we instead establish the ε-robustness of the V-cycle numerically in Section 5; completing this proof is the first direction of future work set out in Section 7.

The Schur-complement preconditioner of Section 3.3 reduces the action of A_δ^{-1} to two inner solves: a pressure mass solve and a solve with the velocity-buoyancy block S. The pressure mass solve is exact: the pressure space L^2_h carries no inter-element continuity, so the mass matrix M_p is block-diagonal with one dense block per element, and a single direct factorisation per element is sufficient.

The Schur complement S is more delicate. Its velocity diagonal $M_u + \frac{\Delta t}{2}C + \frac{\Delta t}{\delta} G M_p^{-1} G^\top$ is a velocity mass term augmented by a grad–div term. An operator of this mass-plus-grad–div form is, up to the parameter weights, the Riesz map of $H(\mathrm{div})$: the operator induced by the inner product $\langle \vec{u}, \vec{v}\rangle + \langle \nabla\cdot\vec{u}, \nabla\cdot\vec{v}\rangle$ that defines the $H(\mathrm{div})$ norm. Arnold, Falk and Winther [1] prove that a multigrid V-cycle with overlapping-Schwarz (patch) smoothing is a uniform preconditioner for this Riesz map, and it is this result that justifies treating S by multigrid. We therefore use a multigrid V-cycle on S, treating it as a single coupled block in $\mathring{H}_h(\mathrm{div}) \times \mathbb{W}_h^\theta$.

Concretely, for each base-mesh vertex v_i the patch ω_i consists of the columns of cells sharing v_i, and X_i is spanned by all velocity-buoyancy degrees of freedom supported on those columns; the corrections from all columns are added with a Richardson damping factor of $\theta = 0.2$. Figure 1 illustrates the resulting degree-of-freedom layout on a representative small-aspect-ratio extruded mesh, with the velocity and buoyancy degrees of freedom on one star patch highlighted.

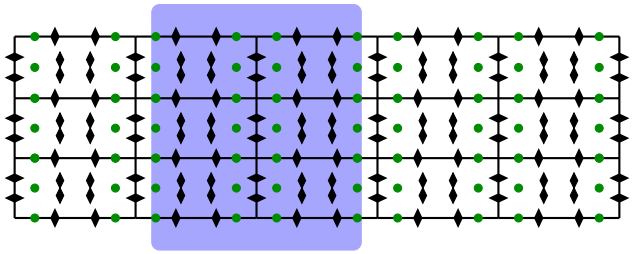


FIG. 1. *Column star patch on a small aspect ratio quadrilateral mesh (`ExtrudedMesh`) for the velocity-buoyancy block obtained after Schur complement elimination of pressure. Black diamonds denote velocity DOFs (RT_2) and green circles denote buoyancy DOFs ($\mathbb{W}_h^\theta = \mathrm{DG}_1 \otimes \mathrm{CG}_2$); the shaded region indicates the column patch subproblem on which the line smoother is applied.*

The column-patch smoother captures the strong vertical coupling induced by the small aspect ratio ε: error modes that are oscillatory along a column are resolved exactly within the patch, while smooth, horizontally-varying modes are deferred to the coarse grid correction. Standard point smoothers (Jacobi, Gauss–Seidel) fail to damp the vertically-oscillatory modes and produce iteration counts that grow as $\varepsilon \to 0$; the column patch restores ε-robustness.

Explicitly, the local problem solved on patch ω_i is: find $(u_i, b_i) \in X_i$ such that

$$s\big(R_i^\top(u_i, b_i),\, R_i^\top(w_i, q_i)\big) = \langle r,\, R_i^\top(w_i, q_i)\rangle \qquad \forall\, (w_i, q_i) \in X_i, \tag{3.29}$$

where r is the current residual of S and $s(\cdot,\cdot)$ is the bilinear form of (3.13); this is precisely the operator equation $A_i(u_i, b_i) = R_i r$ of the abstract construction above, restricted to the velocity and buoyancy degrees of freedom on the columns forming ω_i. Because X_i contains a complete set of vertical-velocity and buoyancy degrees of freedom on those columns together with the horizontal-velocity degrees of freedom on their vertical facets, problem (3.29) is itself a small, well-posed velocity-buoyancy problem and is solved by a direct factorisation; this is the solvability property invoked above and is what makes the patch act as an exact line solve in the vertical direction. The patch smoother is combined with the multigrid hierarchy in the usual subspace-correction sense: relaxation sweeps (3.29) on the fine grid remove the vertically-oscillatory error components patch by patch, and the coarse grid correction, built by the standard geometric coarsening of the extruded mesh in the horizontal direction, removes the remaining smooth, horizontally-varying error. The implementation-level detail of how this patch construction and Richardson relaxation are configured in Firedrake [20] and PETSc [2] for the experiments of Section 5 is given in Appendix B.

3.5. Algorithm summary. The full preconditioner is summarised in Figure 2. At each time step, the fully discrete system (2.17) is solved by FGMRES right-preconditioned by A_δ^{-1}. The preconditioner is applied via the Schur complement factorisation of Section 3.3, with the pressure block inverted exactly by a block-diagonal direct solve and the velocity-buoyancy block inverted approximately by a multigrid V-cycle with column-patch smoothing.

4. Theoretical analysis of the shifted preconditioner. In this section, we establish the rigorous bound on the spectral distance between the shifted operator $\mathcal{A}_\delta$ and the original operator $\mathcal{A}_0$ that motivates the preconditioning strategy of Section 3. The analysis proceeds in three steps. First, we reduce the bound to a well-posedness question for a perturbed mixed variational problem by eliminating the buoyancy vari-

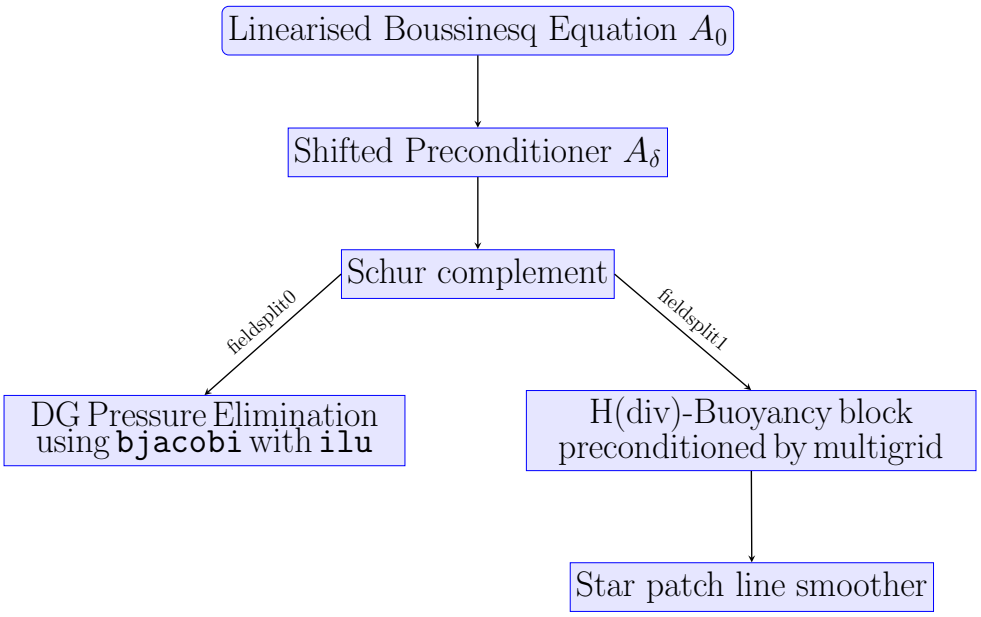


FIG. 2. *Algorithmic structure of the shifted-Schur preconditioner: the fully discrete system A_0 is preconditioned by the shifted matrix A_δ, whose Schur complement splits via* `fieldsplit` *into an exact pressure mass solve (left branch) and a multigrid V-cycle on the velocity-buoyancy block S with column-patch (star) line smoothing (right branch).*

able explicitly. Second, we verify the Brezzi-type conditions on the resulting perturbed problem, obtaining quantitative estimates for the kernel coercivity constant, the inf–sup constant, and the operator norm of the bilinear form. Because the Coriolis term makes the velocity block nonsymmetric, the bound is then drawn from the nonsymmetric perturbed saddle-point theory of [5] rather than from the symmetric perturbed Brezzi theory of [4]; the hypotheses that this requires are coercivity on the constraint kernel alone, which is what the problem supplies. Third, we combine these to state the main bound and derive the asymptotic scaling of the shift parameter δ in terms of the time step Δt, the buoyancy frequency N, and the aspect ratio of the domain.

The analysis is performed at the continuous level. The corresponding discrete statement carries through verbatim provided the compatible finite element triple (2.14) is used, since this triple inherits the inf–sup condition from the continuous setting; we make this explicit at the end of Section 4.3.

4.1. Reduction to a perturbed mixed problem. We work with the fully discrete linearised Boussinesq system (2.17) at a single time step, with the known stage-n data collected into the right-hand sides. For brevity we relabel the unknowns $\vec{u}'^{\,n+1}, b'^{\,n+1}, p'^{\,n+1}$ as u, b, p throughout this section. The system (2.17) can be written as

$$\langle w, u\rangle + \Delta t\langle w, \Omega \times u\rangle - \Delta t\langle \nabla\cdot w, p\rangle - \tfrac{\Delta t}{2}\langle w, b\hat{k}\rangle = \langle f, w\rangle, \tag{4.1a}$$

$$\langle q, b\rangle + \tfrac{\Delta t N^2}{2}\langle u\cdot\hat{k}, q\rangle = \langle g, q\rangle, \tag{4.1b}$$

$$\langle \nabla\cdot u, \phi\rangle = 0, \tag{4.1c}$$

for all test functions $(w, q, \phi) \in V \times B \times Q$, where $V = H(\mathrm{div};\Omega)$ with no-flux essential boundary condition, $B = L^2(\Omega)$, $Q = L^2_0(\Omega)$, and the right-hand sides f, g encode the

stage-n data.

We define the bilinear forms $\mathcal{A}_0, \mathcal{A}_\delta$ on the product space $V \times B \times Q$ by

$$\mathcal{A}_0[u,b,p;\, w,q,\phi] := \langle w, u\rangle + \Delta t\langle w, \Omega \times u\rangle - \Delta t\langle \nabla\cdot w, p\rangle$$

$$- \tfrac{\Delta t}{2}\langle w, b\hat{k}\rangle + \langle q, b\rangle + \tfrac{\Delta t N^2}{2}\langle u\cdot\hat{k}, q\rangle + \langle \nabla\cdot u, \phi\rangle, \tag{4.2}$$

$$\mathcal{A}_\delta[u,b,p;\, w,q,\phi] := \mathcal{A}_0[u,b,p;\, w,q,\phi] + \delta\langle p,\phi\rangle. \tag{4.3}$$

Each of the two forms induces, in the usual way, a linear operator from $V \times B \times Q$ to its dual, and we use the same symbols $\mathcal{A}_0$, $\mathcal{A}_\delta$ for these operators; it is in this sense that compositions such as $\mathcal{A}_\delta^{-1}\mathcal{A}_0$ are understood below. The matrices A_0 and A_δ of Section 3 are the finite element representations of the corresponding discrete variational problems; matrix notation was used there only to display the block structure of the solver, and plays no further role in the analysis, which is carried out at the continuous variational level, the discrete statement being recovered at the end of Section 4.3.

To bound the spectral distance $\|\mathcal{A}_\delta^{-1}\mathcal{A}_0 - I\|$, we set $[u_1,b_1,p_1] := (\mathcal{A}_\delta^{-1}\mathcal{A}_0 - I)[u_0,b_0,p_0]$ for arbitrary input $[u_0,b_0,p_0] \in V\times B\times Q$. Applying $\mathcal{A}_\delta$ to both sides and using $\mathcal{A}_0 - \mathcal{A}_\delta = -\delta\langle p,\phi\rangle$ yields the perturbed problem

$$\langle w, u_1\rangle + \Delta t\langle w, \Omega\times u_1\rangle - \Delta t\langle \nabla\cdot w, p_1\rangle - \tfrac{\Delta t}{2}\langle w, b_1\hat{k}\rangle = 0, \tag{4.4a}$$

$$\langle q, b_1\rangle + \tfrac{\Delta t N^2}{2}\langle u_1\cdot\hat{k}, q\rangle = 0, \tag{4.4b}$$

$$\langle \nabla\cdot u_1, \phi\rangle + \delta\langle p_1,\phi\rangle = -\delta\langle p_0,\phi\rangle, \tag{4.4c}$$

for all $(w,q,\phi) \in V\times B\times Q$. The bound on $\|\mathcal{A}_\delta^{-1}\mathcal{A}_0 - I\|$ is thus reduced to a well-posedness question for (4.4): a bound on $\|(u_1,p_1)\|$ in terms of $\delta\|p_0\|$ translates directly into a bound on the operator norm via supremum over (u_0,b_0,p_0).

We eliminate b_1 from (4.4b) by setting $q = b_1$, which gives $\langle b_1, q\rangle = -\frac{\Delta t N^2}{2}\langle u_1\cdot\hat{k}, q\rangle$ for all $q \in B$. Substituting into (4.4a) via $\frac{\Delta t}{2}\langle w, b_1\hat{k}\rangle = \frac{\Delta t}{2}\langle w\cdot\hat{k}, b_1\rangle = -\frac{\Delta t^2 N^2}{4}\langle w\cdot\hat{k}, u_1\cdot\hat{k}\rangle$ gives, after dropping the subscript on u,p for readability,

$$\left\langle \tfrac{M}{\Delta t}u, w\right\rangle + \langle \Omega\times u, w\rangle - \langle \nabla\cdot w, p\rangle = 0, \tag{4.5a}$$

$$-\langle \nabla\cdot u,\phi\rangle - \delta\langle p,\phi\rangle = \delta\langle p_0,\phi\rangle, \tag{4.5b}$$

where we have rescaled by Δt for symmetry and where

$$M := I + \frac{\Delta t^2 N^2}{4}\,\hat{k}\otimes\hat{k}. \tag{4.6}$$

The operator M is symmetric positive definite, with eigenvalues 1 (on the horizontal components) and $1 + \frac{\Delta t^2 N^2}{4}$ (on the vertical component).

The system (4.5) is of the form

$$a(u,w) + b(w,p) = F[w], \tag{4.7}$$

$$b(u,\phi) - c(p,\phi) = G[\phi], \tag{4.8}$$

with

$$a(u,w) = \left\langle \tfrac{M}{\Delta t}u, w\right\rangle + \langle \Omega\times u, w\rangle, \tag{4.9}$$

$$b(w,p) = -\langle \nabla\cdot w, p\rangle, \tag{4.10}$$

$$c(p,\phi) = \delta\langle p,\phi\rangle, \tag{4.11}$$

$$F[w] = 0, \qquad G[\phi] = \delta\langle p_0,\phi\rangle, \tag{4.12}$$

the sign of G following from (4.4c) on multiplication by -1. Only $\|G\| \le \delta\|p_0\|$ enters the estimates below, so the sign is immaterial to the bound. This is a perturbed saddle point problem of the kind treated in [4] for symmetric a and in [5] without a symmetry assumption; both provide quantitative well-posedness of mixed problems carrying a lower-right perturbation $c(\cdot,\cdot)$. The Coriolis term makes a nonsymmetric here, so it is the latter that we use; see Section 4.3.

Remark 4.1. The elimination of b_1 is exact at the continuous level, holding in the L^2 sense. At the discrete level it holds whenever the compatible finite element triple (2.14) is used and the boundary and the mesh layers are flat, since the vertical-component test space and the buoyancy space agree on horizontal facets. This is the structural reason why the discrete inf–sup constant inherits the continuous estimate, and is the key advantage of the tensor-product buoyancy space $\mathbb{W}_h^\theta$ of [27, 29] over a fully discontinuous alternative.

4.2. Brezzi conditions. We verify the hypotheses required by the abstract perturbed saddle-point theory of Section 4.3 on problem (4.5). Two choices are made before the estimates begin, and both are made in order to expose the aspect-ratio and time-step dependence of the constants rather than to bury it inside them: the domain is rescaled in the vertical, and the velocity space is given a norm weighted by the stratification operator M.

Throughout, the analysis is carried out in rescaled variables, with horizontal lengths measured in units of the horizontal extent of Ω, times in units of a reference time, and the vertical coordinate rescaled by $\hat{z} = z/\varepsilon$, where ε is the aspect ratio. Here z is already the non-dimensional vertical coordinate, measured, like the horizontal lengths, in units of L_x, so that z ranges over $(0,\varepsilon)$ and $\hat{z}$ over $(0,1)$; equivalently, $\hat{z}$ is the depth-normalised coordinate z_{phys}/L_z. The rescaled domain $\hat{\Omega}$ then has unit extent in every coordinate direction and is independent of ε. All of the anisotropy is carried instead by the divergence operator, which becomes

$$\nabla_\varepsilon \cdot \vec{v} := \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{1}{\varepsilon}\frac{\partial v_3}{\partial \hat{z}}. \tag{4.13}$$

This is the same operator as $\nabla\cdot$ written in the original coordinates, so no equation changes. What changes is that the auxiliary problems used to estimate the inf–sup constant in Lemma 4.4 are now posed on a fixed domain rather than on one that degenerates as $\varepsilon \to 0$, and the ε dependence of the constants can be read off directly. We write $\Delta_\varepsilon := \nabla_\varepsilon \cdot \nabla_\varepsilon$ for the correspondingly rescaled Laplacian.

For the spaces, we take

$$V = \mathring{H}(\mathrm{div}_\varepsilon; \hat{\Omega}), \qquad Q = L^2_0(\hat{\Omega}), \tag{4.14}$$

where the circle denotes the no-flux essential condition $\vec{v}\cdot\hat{n} = 0$ and L^2_0 is the space of square-integrable functions of zero mean. Restricting Q to L^2_0 removes the constant pressure mode, which is the physical nullspace of the unshifted operator $\mathcal{A}_0$ and on which no bound is asserted. The velocity space carries the *M-weighted graph norm*

$$\|\vec{v}\|_V^2 := \frac{1}{\Delta t}\,\langle M\vec{v},\, \vec{v}\rangle + \|\nabla_\varepsilon \cdot \vec{v}\|_{L^2}^2, \tag{4.15}$$

with M as in (4.6), and Q carries the L^2 norm. Weighting by $M/\Delta t$ rather than using the plain graph norm $\|\vec{v}\|_{L^2}^2 + \|\nabla_\varepsilon \cdot \vec{v}\|_{L^2}^2$ absorbs the stratification into the metric. Fitting the norm to the operator in this way, so that the parameter dependence is

carried by the metric rather than by the constants, is the strategy of Zulehner [32] for obtaining parameter-robust estimates for saddle point problems, and it is what makes the constants of Lemmas 4.2 and 4.8 independent of Δt and N.

It is convenient to abbreviate

$$m := \lambda_{\max}(M) = 1 + \frac{\Delta t^2 N^2}{4}, \qquad J\vec{v} := \Omega \times \vec{v}, \tag{4.16}$$

so that the form (4.9) reads $a(\vec{u},\vec{w}) = \frac{1}{\Delta t}\langle M\vec{u},\vec{w}\rangle + \langle J\vec{u},\vec{w}\rangle$. On the f-plane $\Omega = (0,0,f/2)$, so that $2\Omega \times \vec{u} = f\,\hat{k} \times \vec{u}$ recovers the Coriolis term of Section 2.1 with f the Coriolis parameter, and $\|J\|_2 = f/2$.

The kernel associated with the bilinear form b is

$$K := \{\vec{v} \in V \ : \ b(\vec{v},q) = 0 \ \text{ for all } q \in Q\}. \tag{4.17}$$

Since $Q = L^2_0$ excludes constants, membership of K says only that $\nabla_\varepsilon \cdot \vec{v}$ is zero. Hence

$$K = \{\vec{v} \in V \ : \ \nabla_\varepsilon \cdot \vec{v} = 0\}, \tag{4.18}$$

which is precisely the subspace on which the norm (4.15) collapses to its first term. Coercivity will be required on K alone and not on all of V.

LEMMA 4.2 (Kernel coercivity and the constraint norms). *With respect to the norm (4.15), the bilinear form a of (4.9) satisfies*

$$a(\vec{v},\vec{v}) = \|\vec{v}\|_V^2 \qquad \forall\, \vec{v} \in K, \tag{4.19}$$

so that the coercivity constant is $\alpha = 1$, exactly, with no dependence on Δt, N, f or the aspect ratio ε. Moreover $\|b\| = 1$ and $\|c\| = \delta$.

Proof. For $\vec{v} \in K$ we have $\nabla_\varepsilon \cdot \vec{v} = 0$ by (4.18), so that $\|\vec{v}\|_V^2 = \frac{1}{\Delta t}\langle M\vec{v},\vec{v}\rangle$. The Coriolis operator is skew, $\langle J\vec{v},\vec{v}\rangle = 0$ pointwise by antisymmetry of the cross product, so it contributes nothing to the quadratic form and

$$a(\vec{v},\vec{v}) = \tfrac{1}{\Delta t}\langle M\vec{v},\vec{v}\rangle + \langle J\vec{v},\vec{v}\rangle = \tfrac{1}{\Delta t}\langle M\vec{v},\vec{v}\rangle = \|\vec{v}\|_V^2. \tag{4.20}$$

This is an equality, not a lower bound, so $\alpha = 1$ is attained. For b, the graph norm structure of (4.15) gives $\|\nabla_\varepsilon \cdot \vec{v}\|_{L^2} \le \|\vec{v}\|_V$ and hence $|b(\vec{v},q)| \le \|\vec{v}\|_V \|q\|_Q$, so $\|b\| \le 1$. Finally $\|c\| = \delta$ is immediate from $c(p,\phi) = \delta\langle p,\phi\rangle$. ∎

The inf–sup condition on b is the key result of this section. Since $b(\vec{v},q)$ depends on $\vec{v}$ only through $\nabla_\varepsilon \cdot \vec{v}$, bounding its supremum from below amounts to exhibiting, for each q, a velocity field whose divergence is $-q$ and whose V-norm is as small as possible. We therefore construct that field first, and establish that the constants it defines are finite; the inf–sup estimate is then immediate.

LEMMA 4.3 (Right inverse of the rescaled divergence). *Let $\hat{\Omega}$ be connected with Lipschitz boundary. For every $q \in Q = L^2_0(\hat{\Omega})$ the auxiliary Neumann problem*

$$-\Delta_\varepsilon \phi = q \ \ \text{in } \hat{\Omega}, \qquad \nabla_\varepsilon \phi \cdot \hat{n} = 0 \ \ \text{on } \partial\hat{\Omega}, \qquad \int_{\hat{\Omega}} \phi = 0, \tag{4.21}$$

has a unique solution, and $\mathcal{R}q := \nabla_\varepsilon \phi$ defines a linear map $\mathcal{R} : Q \to V$ with the following properties.

(i) Right inverse. $\mathcal{R}q \in V$ *and* $\nabla_\varepsilon \cdot \mathcal{R}q = -q$.
(ii) Minimality. $\|\mathcal{R}q\|_{L^2} \le \|\vec{v}\|_{L^2}$ *for every* $\vec{v} \in V$ *with* $\nabla_\varepsilon \cdot \vec{v} = -q$.
(iii) Finiteness. *The two right-inverse constants*

$$C_P := \sup_{q\in Q,\, q\neq 0} \frac{\|\mathcal{R}q\|_{L^2}}{\|q\|_{L^2}}, \qquad C_M := \sup_{q\in Q,\, q\neq 0} \frac{\langle M\mathcal{R}q,\, \mathcal{R}q\rangle^{1/2}}{\|q\|_{L^2}} \tag{4.22}$$

are finite, and

$$C_P = \nu_{\min}^{-1/2}, \qquad C_M \le \sqrt{m}\, C_P, \tag{4.23}$$

where $\nu_{\min} > 0$ *is the smallest nonzero eigenvalue of* $-\Delta_\varepsilon$ *on* $\hat{\Omega}$ *under the boundary conditions of* (4.21).

Here C_P *measures, in* L^2*, how large a velocity field must be in order to realise a prescribed divergence; it is a property of the geometry alone. The constant* C_M *measures the same field in the stratification-weighted metric* $\langle M\cdot,\cdot\rangle$*, and it is* C_M *that enters the estimates below, because* $\|\cdot\|_V$ *weights the velocity by* $M/\Delta t$.

Proof. Existence and uniqueness. The weak form of (4.21) is: find $\phi \in H^1(\hat{\Omega}) \cap L^2_0$ with $\langle \nabla_\varepsilon \phi, \nabla_\varepsilon \psi\rangle = \langle q, \psi\rangle$ for all $\psi \in H^1(\hat{\Omega}) \cap L^2_0$. The form is bounded, and coercive on $H^1 \cap L^2_0$ by the Poincaré–Wirtinger inequality

$$\|\psi\|_{L^2} \;\le\; \nu_{\min}^{-1/2}\, \|\nabla_\varepsilon \psi\|_{L^2} \qquad \forall\, \psi \in H^1(\hat{\Omega}) \cap L^2_0, \tag{4.24}$$

which holds with $\nu_{\min} > 0$ because $\hat{\Omega}$ is connected and the embedding $H^1(\hat{\Omega}) \hookrightarrow L^2(\hat{\Omega})$ is compact. The constraint $q \in L^2_0$ is exactly the compatibility condition for the Neumann problem, so Lax–Milgram gives a unique ϕ.

(i). By construction $\mathcal{R}q = \nabla_\varepsilon \phi \in L^2(\hat{\Omega})$ and $\nabla_\varepsilon \cdot \mathcal{R}q = \Delta_\varepsilon \phi = -q \in L^2(\hat{\Omega})$, so $\mathcal{R}q \in H(\mathrm{div}_\varepsilon; \hat{\Omega})$. Its normal trace is $\mathcal{R}q \cdot \hat{n} = \nabla_\varepsilon \phi \cdot \hat{n} = 0$, which is the no-flux condition, so $\mathcal{R}q \in \mathring{H}(\mathrm{div}_\varepsilon) = V$.

(ii). Let $\vec{v} \in V$ with $\nabla_\varepsilon \cdot \vec{v} = -q$, and put $\vec{v}_0 := \vec{v} - \mathcal{R}q$, so that $\nabla_\varepsilon \cdot \vec{v}_0 = 0$ and $\vec{v}_0 \cdot \hat{n} = 0$. Integrating by parts,

$$\langle \mathcal{R}q,\, \vec{v}_0\rangle = \langle \nabla_\varepsilon \phi,\, \vec{v}_0\rangle = -\,\langle \phi,\, \nabla_\varepsilon \cdot \vec{v}_0\rangle + \int_{\partial\hat{\Omega}} \phi\, \vec{v}_0 \cdot \hat{n} = 0, \tag{4.25}$$

both terms vanishing, the first because $\vec{v}_0$ is divergence-free and the second because $\vec{v}_0 \in V$. Hence $\|\vec{v}\|^2_{L^2} = \|\mathcal{R}q\|^2_{L^2} + \|\vec{v}_0\|^2_{L^2} \ge \|\mathcal{R}q\|^2_{L^2}$.

(iii). Let $\{\psi_k\}$ be an L^2-orthonormal basis of eigenfunctions of $-\Delta_\varepsilon$ under the boundary conditions of (4.21), with eigenvalues ν_k; the zero-mean constraint excludes the eigenvalue 0, so $\nu_k \ge \nu_{\min} > 0$. Integration by parts gives $\langle \nabla_\varepsilon \psi_j, \nabla_\varepsilon \psi_k\rangle = \nu_k \delta_{jk}$. For $q = \sum_k c_k \psi_k$ we have $\phi = \sum_k \nu_k^{-1} c_k \psi_k$, hence

$$\frac{\|\mathcal{R}q\|^2_{L^2}}{\|q\|^2_{L^2}} = \frac{\sum_k \nu_k^{-2} c_k^2\, \nu_k}{\sum_k c_k^2} = \frac{\sum_k c_k^2/\nu_k}{\sum_k c_k^2} \;\le\; \frac{1}{\nu_{\min}}, \tag{4.26}$$

with equality when q is an eigenfunction for $\nu_{\min}$. This gives $C_P = \nu_{\min}^{-1/2}$, finite by (4.24). Finally $\langle M\vec{w}, \vec{w}\rangle \le m\|\vec{w}\|^2_{L^2}$ by (4.16), so $C_M \le \sqrt{m}\, C_P < \infty$. ∎

Lemma 4.4 (Inf–sup constant of $b(\cdot,\cdot)$). *With* $\mathcal{R}$*,* C_M *and* C_P *as in Lemma 4.3, the inf–sup condition holds:*

$$\inf_{q\in Q} \sup_{\vec{v}\in V} \frac{b(\vec{v}, q)}{\|q\|_Q \|\vec{v}\|_V} \;\ge\; \beta := \left(1 + \frac{C_M^2}{\Delta t}\right)^{-1/2} \;\ge\; \left(1 + \frac{m\, C_P^2}{\Delta t}\right)^{-1/2} \;>\; 0. \tag{4.27}$$

Proof. Fix $q \in Q$ and take $\vec{v} := \mathcal{R}q$, which lies in V by Lemma 4.3(i). Then

$$b(\vec{v}, q) = -\langle \nabla_\varepsilon \cdot \vec{v},\, q\rangle = \|q\|_{L^2}^2. \tag{4.28}$$

Using $\nabla_\varepsilon \cdot \vec{v} = -q$ in the second term of (4.15) and the definition (4.22) of C_M in the first,

$$\|\vec{v}\|_V^2 = \frac{1}{\Delta t}\,\langle M\mathcal{R}q,\, \mathcal{R}q\rangle + \|q\|_{L^2}^2 \le \left(1 + \frac{C_M^2}{\Delta t}\right)\|q\|_{L^2}^2, \tag{4.29}$$

so that $b(\vec{v}, q)/\|\vec{v}\|_V \ge \beta\|q\|_Q$. Taking the supremum over $\vec{v} \in V$ and then the infimum over $q \in Q$ gives the first inequality; the second follows from $C_M \le \sqrt{m}\, C_P$. Both are strictly positive because C_M is finite, by Lemma 4.3(iii). ∎

We now make the ε-dependence of these constants explicit. The inf–sup constant of Lemma 4.4 depends on the geometry only through C_M, and on a product domain the Poincaré constants split into a horizontal and a vertical contribution, with all of the ε-dependence carried by the vertical one.

LEMMA 4.5 (Horizontal–vertical splitting of the Poincaré constant). *Suppose the rescaled domain is a product, $\hat{\Omega} = \omega \times (0,1)$, with ω a connected Lipschitz horizontal domain carrying periodic or homogeneous Neumann (free-slip) lateral conditions, and with the Neumann conditions of (4.21) at $\hat{z} \in \{0,1\}$. Let $A_1 > 0$ be the smallest nonzero eigenvalue of the horizontal Laplacian $-\Delta_h$ on ω, and let $\lambda_1 > 0$ be the smallest nonzero eigenvalue of the vertical operator $-\partial^2/\partial\hat{z}^2$ on the unit interval $(0,1)$ under the top and bottom conditions of (4.21). Define the horizontal and vertical Poincaré constants*

$$C_h := A_1^{-1/2}, \qquad C_v := \varepsilon\,\hat{C}_v, \qquad \hat{C}_v := \lambda_1^{-1/2}, \tag{4.30}$$

so that C_v is the Poincaré constant of the rescaled vertical operator $-\varepsilon^{-2}\,\partial^2/\partial\hat{z}^2$ on $(0,1)$. Since ω is the rescaled horizontal domain and $\hat{C}_v$ is a constant of the unit interval, both C_h and $\hat{C}_v$ are independent of ε, and all of the ε-dependence is carried by C_v. For the rigid-lid (Neumann) conditions used throughout this report, $\lambda_1 = \pi^2$, so $\hat{C}_v = 1/\pi$ and $C_v = \varepsilon/\pi$. Then

$$C_P = \max(C_h,\, C_v), \qquad C_M^2 = \max\!\left(C_h^2,\; m\,C_v^2\right). \tag{4.31}$$

In particular, whenever $\sqrt{m}\,C_v \le C_h$, we have $C_M = C_P = C_h$, with no dependence on ε, Δt or N.

Proof. On the product domain the eigenfunctions of $-\Delta_\varepsilon$ under the stated boundary conditions separate: they are products $\psi_{jl} = \phi_j\, v_l$ of eigenfunctions ϕ_j of $-\Delta_h$ on ω (eigenvalues A_j listed in increasing order with multiplicity, $A_0 = 0$ for the constant mode, so that A_1 is the smallest nonzero eigenvalue of the statement) and eigenfunctions v_l of the vertical operator on $(0,1)$ (eigenvalues $B_l = \lambda_l/\varepsilon^2$ with $\lambda_0 = 0 < \lambda_1 \le \cdots$, the factor $1/\varepsilon^2$ arising from the vertical part of (4.13); for the rigid-lid conditions these are the Neumann modes $v_l(\hat{z}) = \cos(\pi l\hat{z})$ with $\lambda_l = \pi^2 l^2$), so that $\nu_{jl} = A_j + B_l$ are the eigenvalues. The zero-mean constraint excludes $(j,l) = (0,0)$. Normalise $\|\psi_{jl}\|_{L^2} = 1$ and take $q = \psi_{jl}$. The auxiliary solution of (4.21) is then $\phi = \nu_{jl}^{-1}\psi_{jl}$, so $\mathcal{R}q = \nu_{jl}^{-1}\nabla_\varepsilon\psi_{jl}$, whose horizontal part is $\nu_{jl}^{-1}(\nabla_h\phi_j)\,v_l$ and whose vertical part is $\nu_{jl}^{-1}\varepsilon^{-1}\phi_j\,v_l'\,\hat{k}$. Integrating by parts under the stated boundary conditions,

$$\|\nabla_h\psi_{jl}\|_{L^2}^2 = \langle -\Delta_h\psi_{jl},\, \psi_{jl}\rangle = A_j, \qquad \|\varepsilon^{-1}\partial_{\hat{z}}\psi_{jl}\|_{L^2}^2 = \left\langle -\varepsilon^{-2}\partial_{\hat{z}\hat{z}}\psi_{jl},\, \psi_{jl}\right\rangle = B_l, \tag{4.32}$$

so the horizontal and vertical parts of $\nabla_\varepsilon \psi_{jl}$ carry A_j and B_l respectively, and

$$\|\mathcal{R}q\|_{L^2}^2 = \nu_{jl}^{-2}\,(A_j + B_l) = \frac{1}{\nu_{jl}}. \tag{4.33}$$

Since $M = I + \frac{\Delta t^2 N^2}{4}\,\hat{k}\otimes\hat{k}$ acts as the identity on the horizontal components and multiplies the vertical component by m,

$$\langle M\mathcal{R}q,\,\mathcal{R}q\rangle = \frac{1}{\nu_{jl}^2}\,\langle M\nabla_\varepsilon\psi_{jl},\,\nabla_\varepsilon\psi_{jl}\rangle = \frac{1}{\nu_{jl}^2}\left(\|\nabla_h\psi_{jl}\|_{L^2}^2 + m\,\|\varepsilon^{-1}\partial_{\hat{z}}\psi_{jl}\|_{L^2}^2\right) = \frac{A_j + mB_l}{\nu_{jl}^2}. \tag{4.34}$$

Hence

$$\frac{\|\mathcal{R}q\|_{L^2}^2}{\|q\|_{L^2}^2} = \frac{1}{\nu_{jl}}, \qquad \frac{\langle M\mathcal{R}q,\,\mathcal{R}q\rangle}{\|q\|_{L^2}^2} = g(A_j, B_l), \qquad g(A,B) := \frac{A+mB}{(A+B)^2}. \tag{4.35}$$

Both quantities are diagonal in this basis, so each supremum in (4.22) is attained on a single mode.

The first is maximised by minimising ν_{jl}: every mixed mode has $\nu_{jl} \geq A_1 + B_1$, so $\nu_{\min} = \min(A_1, B_1)$ and $C_P^2 = \max(A_1^{-1},\, B_1^{-1}) = \max(C_h^2,\, C_v^2)$.

For the second, note that

$$\frac{\partial g}{\partial A} = \frac{B(1-2m) - A}{(A+B)^3} < 0 \qquad \text{for } A, B \geq 0 \text{ not both zero}, \tag{4.36}$$

since $m \geq 1$. Thus g is strictly decreasing in A, so for any $B > 0$ the best choice is $A = 0$, and among those the best is the smallest admissible B, namely B_1. The only remaining candidates have $B = 0$, where $g = 1/A$ is maximised by the smallest admissible A, namely A_1. Hence

$$C_M^2 = \max\big(g(A_1, 0),\; g(0, B_1)\big) = \max\left(\frac{1}{A_1},\; \frac{m}{B_1}\right) = \max\big(C_h^2,\; m\,C_v^2\big)\,. \tag{4.37}$$

Finally, if $\sqrt{m}\,C_v \leq C_h$ then also $C_v \leq C_h$, since $m \geq 1$, so the horizontal branch attains both maxima and $C_M = C_P = C_h$. ∎

The specific values on the rectangular domain of Section 2.1 follow as a special case.

Corollary 4.6 (Aspect-ratio robustness). *Let $\hat{\Omega}$ be the rescaled box of unit extent in every coordinate direction, periodic in the horizontal directions and with no-flux top and bottom, obtained either from the two-dimensional slice domain $\Omega = (0, L_x)\times(0, L_z)$ of Section 2.1 or from the full three-dimensional domain $\Omega = (0, L_x)\times(0, L_y)\times(0, L_z)$ from which the slice is obtained by y-independence. Write $\varepsilon = L_z/L_x$. Then, for every $\varepsilon \leq \frac{1}{2}$, the constants of (4.22) are*

$$C_P = \frac{1}{2\pi}, \qquad C_M^2 = C_P^2\,\max\big(1,\; 4\varepsilon^2 m\big) = \frac{1}{4\pi^2}\,\max\Big(1,\; 4\varepsilon^2\Big(1 + \tfrac{\Delta t^2 N^2}{4}\Big)\Big)\,, \tag{4.38}$$

in two and in three dimensions alike. In particular, whenever $2\varepsilon\sqrt{m} \leq 1$ — equivalently $\varepsilon\,\Delta t\,N \leq \sqrt{1-4\varepsilon^2}$, which for a thin domain is simply $\varepsilon\,\Delta t\,N \lesssim 1$ — we have $C_M = C_P = 1/(2\pi)$.

Proof. The box is a product domain of the form required by Lemma 4.5: in the slice, ω is the unit periodic interval, whose eigenvalues are $A_j = 4\pi^2 j^2$; in three dimensions ω is the unit periodic square, whose additional horizontal index contributes further multiples of $4\pi^2$ and does not alter $A_1 = 4\pi^2$. Hence $C_h = 1/(2\pi)$ and, the lids being rigid, $\hat{C}_v = 1/\pi$ and $C_v = \varepsilon/\pi$; then (4.31) gives (4.38): the condition $\varepsilon \le \frac{1}{2}$ is exactly $C_v \le C_h$, and $m\,C_v^2 \le C_h^2$ is exactly $4\varepsilon^2 m \le 1$, that is $2\varepsilon\sqrt{m} \le 1$. ∎

The same computation can be read in physical units, which is the form the numerical experiments of Section 5 use.

COROLLARY 4.7 (Dimensional form of the constants). *In physical units, with lengths no longer measured in units of L_x, the constants of Corollary 4.6 read*

(4.39)
$$C_h = \frac{L_x}{2\pi}, \qquad C_v = \frac{L_z}{\pi}, \qquad C_P = \max(C_h,\, C_v), \qquad C_M^2 = \max\bigl(C_h^2,\; m\,C_v^2\bigr) :$$

the horizontal constant is set by the horizontal extent, the vertical constant by the depth, and $C_M = C_P = L_x/(2\pi)$ whenever $2\varepsilon\sqrt{m} \le 1$, that is throughout the thin-domain regime $\varepsilon\,\Delta t\,N \lesssim 1$.

Proof. Undoing the length normalisation multiplies each constant carrying a dimension of length by L_x: $C_h = L_x/(2\pi)$ and $C_v = \varepsilon L_x \hat{C}_v = L_z/\pi$. The maxima of Lemma 4.5 are unchanged, since m, ε and the condition $2\varepsilon\sqrt{m} \le 1$ are dimensionless. ∎

LEMMA 4.8 (Operator norm of a). *With respect to the norm (4.15), the bilinear form a defined by (4.9) has operator norm*

$$\|a\| = \sqrt{1 + \frac{\Delta t^2 f^2}{4}}, \tag{4.40}$$

and the value is attained.

Proof. Write $a(\vec{v},\vec{w}) = \langle A\vec{v},\vec{w}\rangle$ with $A = \Delta t^{-1}M + J$, and $\|\vec{v}\|_V^2 = \langle G\vec{v},\vec{v}\rangle$ with $G = \Delta t^{-1}M + D^*D$, $D := \nabla_\varepsilon\cdot$. By Cauchy–Schwarz in the G-inner product, $\|a\| = \|G^{-1/2}AG^{-1/2}\|_2 \le \|\hat{A}\|_2$, where

$$\hat{A} := \bigl(\Delta t^{-1}M\bigr)^{-1/2} A \bigl(\Delta t^{-1}M\bigr)^{-1/2} = I + \Delta t\, M^{-1/2} J M^{-1/2}, \tag{4.41}$$

the inequality holding because $G \ge \Delta t^{-1}M$. Since $M = \mathrm{diag}(1,1,m)$ with $m = 1 + \Delta t^2 N^2/4$, and J acts only on the horizontal components, $M^{-1/2}$ is the identity where J is supported and cancels the vertical entry $m/\Delta t$ of A exactly, so

$$\hat{A} = \begin{pmatrix} 1 & -\frac{\Delta t f}{2} & 0 \\ \frac{\Delta t f}{2} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \hat{A}_h^{\top}\hat{A}_h = \Bigl(1 + \tfrac{\Delta t^2 f^2}{4}\Bigr) I_2, \tag{4.42}$$

the cross terms in $\hat{A}_h^{\top}\hat{A}_h$ cancelling because the horizontal block of J is skew. As $\hat{A}$ is block diagonal, $\|\hat{A}\|_2 = \max\{\sqrt{1+\Delta t^2 f^2/4},\, 1\} = \sqrt{1+\Delta t^2 f^2/4}$, which gives (4.40) as an upper bound.

For attainment, let $\vec{v} \in K$ be any nonzero horizontal field with $\nabla_\varepsilon\cdot\vec{v} = 0$ and set $\vec{w} := (I + \Delta t J)\vec{v}$. Both fields are horizontal, so M acts as the identity on them and $\|\cdot\|_V^2$ reduces to $\Delta t^{-1}\|\cdot\|_{L^2}^2$; evaluating directly, $a(\vec{v},\vec{w})/\left(\|\vec{v}\|_V\|\vec{w}\|_V\right) = \sqrt{1+\Delta t^2 f^2/4}$. ∎

Remark 4.9. Rotation is the only source of nonsymmetry in a; being skew, it contributes nothing to the coercivity constant (Lemma 4.2) and enters the analysis solely through $\|a\|$, and there only at second order in the dimensionless group $\Delta t f$ (Lemma 4.8). For the mid-latitude value $f \approx 10^{-4}$ s^{-1} and any usable time step, $\Delta t f/2 \lesssim 10^{-2}$, so $\|a\| - 1 = \frac{1}{2}(\Delta t f/2)^2 \lesssim 5 \times 10^{-5}$: rotation is numerically invisible in the bound. A numerical sweep in the Coriolis parameter should accordingly be reported against $\Delta t f$ rather than against f alone.

4.3. The main bound. We give two bounds. The first assumes $f = 0$, so that a is symmetric; it is proved directly, from the pressure Schur complement, and is sharp. The second makes no symmetry assumption and is obtained from the abstract nonsymmetric theory; it is necessarily looser. The two agree exactly at leading order in the small parameter δ/β^2 and differ only at second order (Section 4.5), which is a useful cross-check on both.

When $J = 0$ the form $a(\vec{v}, \vec{w}) = \frac{1}{\Delta t}\langle M\vec{v}, \vec{w}\rangle$ is symmetric and positive definite. The perturbed problem then falls within the classical symmetric perturbed theory, and no result beyond Brezzi's Theorem 4.3.2 in [4, Section 4.3] is required.

THEOREM 4.10 (Convergence bound, symmetric case). *Let $f = 0$, so that a is symmetric, and let $\mathcal{A}_0$ and $\mathcal{A}_\delta$ be the operators defined in (4.2)–(4.3) associated with the linearised Boussinesq system (4.1) on the rescaled domain $\hat{\Omega}$, either the three-dimensional domain of Section 2.1 or its two-dimensional vertical-slice specialisation, as in Corollary 4.6. Then, for every $\delta > 0$,*

$$\left\| \mathcal{A}_\delta^{-1}\mathcal{A}_0 - I \right\|_{V\times Q} \;\le\; (C_1 + C_2)\,\delta, \tag{4.43}$$

where the product norm on $V \times Q$ is $\|(\vec{u}, p)\|_{V\times Q} := \|\vec{u}\|_V + \|p\|_Q$ and

$$C_1 = \frac{2}{\beta} = 2\sqrt{1 + \frac{C_M^2}{\Delta t}}, \tag{4.44}$$

$$C_2 = \frac{4}{\delta + 2\beta^2} = \frac{4\left(\Delta t + C_M^2\right)}{\delta\left(\Delta t + C_M^2\right) + 2\,\Delta t}, \tag{4.45}$$

with β and C_M as in Lemmas 4.3–4.4.

Proof. With $f = 0$ the form a is symmetric, so Brezzi's Theorem 4.3.2 in [4] applies to the perturbed problem (4.5) directly, with the constants of Lemmas 4.2, 4.4 and 4.8. It yields the separate bounds

$$\|\vec{u}\|_V \le \left(\frac{\beta^2 + 4\delta\|a\|}{\alpha\,\beta^2}\right)\|F\|_{V'} + \frac{2\|a\|^{1/2}}{\alpha^{1/2}\,\beta}\,\|G\|_{Q'}, \tag{4.46}$$

$$\|p\|_Q \le \frac{2\|a\|^{1/2}}{\alpha^{1/2}\,\beta}\,\|F\|_{V'} + \frac{4\|a\|}{\delta\|a\| + 2\beta^2}\,\|G\|_{Q'}. \tag{4.47}$$

As in the proof of Theorem 4.11 below, applying $\mathcal{A}_\delta$ to the definition of $(\vec{u}, p)$ shows that it solves (4.5) with $F = 0$ and $G = \delta\, p_0$, so that $\|G\|_{Q'} \le \delta\|p_0\|_Q$. Substituting $\alpha = 1$ (Lemma 4.2) and $\|a\| = 1$ (Lemma 4.8 with $f = 0$) leaves

$$\|\vec{u}\|_V \le C_1\,\delta\,\|p_0\|_Q, \qquad \|p\|_Q \le C_2\,\delta\,\|p_0\|_Q, \tag{4.48}$$

with C_1, C_2 as in (4.44)–(4.45), the second expressions following from $\beta^2 = \Delta t/(\Delta t + C_M^2)$. Summing the two bounds and taking the supremum over $(\vec{u}_0, b_0, p_0)$ gives (4.43). ∎

Without the assumption $f = 0$, problem (4.5) is a perturbed saddle-point problem with a *nonsymmetric* velocity block: the Coriolis term makes a nonsymmetric, whereas the classical perturbed Brezzi theory of [4, Section 4.3] is stated for symmetric a. We therefore work from Theorem 3.2 of Ciarlet, Huang and Zou [5], which treats

$$a(u,v) + b_1(v,p) = (F,v)_V, \qquad b_2(u,q) - c(p,q) = (G,q)_Q, \tag{4.49}$$

with possibly distinct off-diagonal forms, with no symmetry assumption on a, and under hypotheses stated on the constraint kernels rather than on the whole velocity space. Specialising to $b_1 = b_2 = b$ collapses the pair of inf–sup conditions to a single one and identifies both kernels with the K of (4.17), leaving three hypotheses, in which $\|a\|$, $\|b\|$ and $\|c\|$ are operator norms with respect to the pair $(\|\cdot\|_V, \|\cdot\|_Q)$:

(A1) *Continuity.* a, b and c are bounded, with $\|c\| = \delta$.
(A2) *Coercivity on the kernel.* $a(\vec{v},\vec{v}) \geq \alpha \|\vec{v}\|_V^2$ for all $\vec{v} \in K$, with $\alpha > 0$.
(A3) *Inf–sup.* $\sup_{\vec{v}\in V} b(\vec{v},q)/\|\vec{v}\|_V \geq \beta \|q\|_Q$ for all $q \in Q$, with $\beta > 0$.

No sign condition on c is required, and since $a(\vec{v},\vec{v})$ depends only on the symmetric part of a, the skew Coriolis part is invisible to (A2) and enters only through $\|a\|$ in (A1). Lemmas 4.2, 4.8 and 4.4 establish exactly (A1)–(A3) for (4.5), with the four constants

$$\alpha = 1, \qquad \|a\| = \sqrt{1 + \tfrac{\Delta t^2 f^2}{4}}, \qquad \|b\| = 1, \qquad \|c\| = \delta, \qquad \beta = \left(1 + \tfrac{C_M^2}{\Delta t}\right)^{-1/2}. \tag{4.50}$$

The single smallness hypothesis of the theorem is that

$$\delta_2 := \frac{\|a\|}{\beta^2}\left(1 + \frac{\|a\|}{\alpha}\right)\delta \;<\; 1. \tag{4.51}$$

THEOREM 4.11 (Convergence bound for the shifted preconditioner). *Let $\mathcal{A}_0$ and $\mathcal{A}_\delta$ be the operators defined in (4.2)–(4.3) associated with the fully discrete linearised f-plane Boussinesq system (4.1) on the rescaled domain $\hat{\Omega}$, either the three-dimensional domain of Section 2.1 or its two-dimensional vertical-slice specialisation, as in Corollary 4.6. Assume (A1)–(A3) with the constants (4.50), and assume $\delta_2 < 1$. Then problem (4.5) has a unique solution; writing $(\vec{u},p) := (\mathcal{A}_\delta^{-1}\mathcal{A}_0 - I)(\vec{u}_0,p_0)$,*

$$\|\vec{u}\|_V \leq \frac{1}{1-\delta_2} \cdot \frac{\delta}{\beta}\left(1 + \frac{\|a\|}{\alpha}\right)\|p_0\|_Q, \tag{4.52}$$

$$\|p\|_Q \leq \frac{\delta_2}{1-\delta_2}\,\|p_0\|_Q, \tag{4.53}$$

and consequently

$$\left\|\mathcal{A}_\delta^{-1}\mathcal{A}_0 - I\right\|_{V\times Q} \;\leq\; \frac{1}{1-\delta_2}\left(1 + \frac{\|a\|}{\alpha}\right)\left(\frac{1}{\beta} + \frac{\|a\|}{\beta^2}\right)\delta, \tag{4.54}$$

where the product norm on $V \times Q$ is $\|(\vec{u},p)\|_{V\times Q} := \|\vec{u}\|_V + \|p\|_Q$. Substituting the constants (4.50), and abbreviating $\|a\| = \sqrt{1 + \Delta t^2 f^2/4}$, this reads

$$\left\|\mathcal{A}_\delta^{-1}\mathcal{A}_0 - I\right\|_{V\times Q} \;\leq\; \frac{1 + \|a\|}{1-\delta_2}\left(\frac{1}{\beta} + \frac{\|a\|}{\beta^2}\right)\delta, \qquad \delta_2 = \frac{\|a\|\,(1+\|a\|)}{\beta^2}\,\delta < 1. \tag{4.55}$$

Proof. Applying $\mathcal{A}_\delta$ to the definition of $(\vec{u}, p)$ and using $\mathcal{A}_0 - \mathcal{A}_\delta = -\delta\langle p, \phi\rangle$ shows, exactly as in (4.4), that $(\vec{u}, p)$ solves the perturbed problem (4.5) with data $F = 0$ and $G = \delta\, p_0$, so that $\|G\|_Q = \delta\|p_0\|_Q$.

Hypotheses (A1)–(A3) hold by Lemmas 4.2, 4.8 and 4.4, and the kernel identification (4.18) identifies the abstract constraint kernel of [5] with the divergence-free subspace. Their Theorem 3.2 phrases its estimates as a comparison between the solution of the perturbed problem and the solution $(\tilde{\vec{u}}, \tilde{p})$ of the corresponding unperturbed problem. The unperturbed stability estimates give

$$\|\tilde{\vec{u}}\|_V \le \frac{1}{\beta}\left(1 + \frac{\|a\|}{\alpha}\right)\delta\,\|p_0\|_Q, \qquad \|\tilde{p}\|_Q \le \frac{\|a\|}{\beta}\,\|\tilde{\vec{u}}\|_V = \delta_2\,\|p_0\|_Q, \tag{4.56}$$

the last equality being the definition (4.51) of δ_2. Substituting these into the estimates of Theorem 3.2 and collecting terms gives (4.52) and (4.53). Adding the two and taking the supremum over $(\vec{u}_0, b_0, p_0)$ gives

$$\left\|\mathcal{A}_\delta^{-1}\mathcal{A}_0 - I\right\|_{V\times Q} = \sup_{(\vec{u}_0,b_0,p_0)\neq 0} \frac{\|(\vec{u},p)\|_{V\times Q}}{\|(\vec{u}_0,p_0)\|_{V\times Q}} \le \frac{1}{1-\delta_2}\left(1 + \frac{\|a\|}{\alpha}\right)\left(\frac{1}{\beta} + \frac{\|a\|}{\beta^2}\right)\delta, \tag{4.57}$$

which is (4.54); inserting $\alpha = 1$ yields (4.55). ∎

In the regime of interest, $\delta \ll \beta^2 \ll 1$, the two bounds agree to leading order, giving the common threshold $\delta \lesssim \beta^2/2$ (the expansion is carried out in Section 4.5). Since the constants come from different theorems proved by different arguments, the agreement is genuine corroboration of both. Theorem 4.11 requires the smallness hypothesis $\delta_2 < 1$ even to be stated, while Theorem 4.10 holds for every $\delta > 0$. In the symmetric case the pressure can be eliminated exactly, and the shifted pressure operator $S_p + \delta I$, with S_p the positive semi-definite pressure Schur complement of the unshifted problem, only improves as δ grows.

Finally, the bound transfers to the discrete setting. Theorem 3.2 of [5] is stated for Hilbert spaces of arbitrary dimension, so Theorem 4.11 holds verbatim on the compatible discrete pair $V_h \times Q_h$ of (2.14) provided the four constants (4.50) transfer, and they do. Continuity is inherited by conformity. For coercivity, the compatible triple gives $\nabla_\varepsilon \cdot V_h \subseteq Q_h$, so a discretely divergence-free field is exactly divergence-free and $\alpha_h = \alpha = 1$; this inclusion is what the discrete de Rham structure of the triple guarantees [7]. The inf–sup constant satisfies $\beta_h \ge \beta$ by the Fortin argument built on the commuting interpolant of the Raviart–Thomas space [4, Chapter 7], and the elimination of b_1 in Remark 4.1 is exact under the choice of buoyancy space $\mathbb{W}_h^\theta$.

4.4. Aspect-ratio robustness. The aspect ratio enters the bound (4.54) only through β, and hence only through C_M. The coercivity constant $\alpha = 1$ (Lemma 4.2) and the norm $\|a\|$ (Lemma 4.8) do not see the geometry. By Lemma 4.5 the ε-dependence of C_M sits entirely in the vertical constant $C_v = \varepsilon/\pi$, and by Corollary 4.6 $C_M = C_h = 1/(2\pi)$ whenever $\varepsilon\,\Delta t\,N \lesssim 1$. In that regime β, and with it the whole bound, is exactly independent of ε. This is the analytical counterpart of the robustness observed in Section 5, and it holds in precisely the limit in which standard preconditioners degrade. The rescaling plays no part in it; in physical units the same computation gives $C_h = L_x/(2\pi)$ and $C_v = L_z/\pi$ (Corollary 4.7), so C_M is set by the longest direction of the domain and does not degenerate as $L_z \to 0$.

The statement concerns the worst pressure mode. The supremum defining C_M is attained on the depth-independent mode $(j, l) = (1, 0)$, whose constant C_h does not

involve L_z; modes with vertical structure carry the vertical branch $mC_v^2 = m\varepsilon^2/\pi^2$ of (4.31), which improves like ε^2. Uniformity in ε therefore guarantees that thinner domains never converge more slowly, while data confined to modes with vertical structure converge faster as the domain thins. Section 5.3 observes exactly this (Remark 5.1).

4.5. Asymptotic shift-parameter scaling. Write $\rho := \|\mathcal{A}_\delta^{-1}\mathcal{A}_0 - I\|_{V\times Q}$ for the quantity bounded in Theorems 4.10 and 4.11. If $\rho < 1$ the preconditioned Richardson iteration contracts the error by the factor ρ at every step, and GMRES, which minimises the residual over a Krylov space containing the Richardson iterates, converges at least as fast; right preconditioning replaces $\mathcal{A}_\delta^{-1}\mathcal{A}_0$ by the similar operator $\mathcal{A}_0\mathcal{A}_\delta^{-1}$ and changes nothing. We therefore ask when $\rho < 1$, and how the admissible δ scales with Δt, N and ε.

For $\delta \ll \beta^2 \ll 1$ the two bounds can be expanded directly. The symmetric bound (4.43) gives

$$\rho \;\le\; (C_1 + C_2)\,\delta = \frac{2\delta}{\beta} + \frac{4\delta}{\delta + 2\beta^2} = \frac{2\delta}{\beta^2} + \frac{2\delta}{\beta} + O(\delta^2) \;\simeq\; \frac{2\delta}{\beta^2}, \tag{4.58}$$

the first term dominating because $\beta \ll 1$. In the general bound (4.55) the only new ingredient is $\|a\| = 1 + O(\Delta t^2 f^2)$, which equals one to the accuracy of Remark 4.9; setting $\|a\| = 1$ there reproduces the same leading term $2\delta/\beta^2$. The two bounds therefore agree to leading order, which corroborates the constants of both, since they are proved by different arguments, and a single analysis covers the symmetric and the general case.

To leading order the condition $\rho < 1$ therefore reads $2\delta/\beta^2 < 1$, that is

$$\delta \;\lesssim\; \frac{\beta^2}{2} = \frac{1}{2}\left(1 + \frac{C_M^2}{\Delta t}\right)^{-1}, \tag{4.59}$$

which in the relevant regime $C_M^2 \gg \Delta t$ is $\delta \lesssim \Delta t/(2C_M^2)$. The exact thresholds differ from this only by constants of order one, which scale the admissible shift but not its dependence on the parameters. That dependence is now concentrated in C_M, which Corollary 4.6 evaluates exactly, and the two branches of the maximum in (4.38) split the condition into two regimes.

PROPOSITION 4.12 (Asymptotic shift-parameter scaling). *Let $\hat{\Omega}$ be the rescaled box of Corollary 4.6, let $N > 0$, assume $C_M^2 \gg \Delta t$. Then the convergence condition is satisfied when*

$$\delta \;\lesssim\; \frac{\Delta t}{2\,C_P^2\,\max(1,\,4\varepsilon^2 m)}, \qquad m = 1 + \frac{\Delta t^2 N^2}{4}, \qquad C_P = \frac{1}{2\pi}, \tag{4.60}$$

which separates into the following two regimes.

1. Weakly stratified regime, $\varepsilon\,\Delta t\,N \lesssim 1$. *Here $C_M = C_P$ and*

$$\delta \;\lesssim\; \frac{\Delta t}{2C_P^2} = 2\pi^2\,\Delta t, \qquad \textit{that is} \qquad \delta \propto \Delta t, \tag{4.61}$$

   with no dependence on N and none on the aspect ratio.

2. Strongly stratified regime, $\varepsilon\,\Delta t\,N \gtrsim 1$. *Here $C_M^2 \simeq C_P^2\,\varepsilon^2\Delta t^2 N^2$ and*

$$\Delta t\,\delta \;\lesssim\; \frac{1}{2\,C_P^2\,\varepsilon^2 N^2} = \frac{2\pi^2}{\varepsilon^2 N^2}, \qquad \textit{that is} \qquad \delta \propto \frac{1}{\Delta t}. \tag{4.62}$$

The two regimes meet at $\Delta t \sim 1/(\varepsilon N)$, in the symmetric and the general case alike: the leading constant scales the admissible shift but does not move the cross-over, which is determined by $\max(1, 4\varepsilon^2 m)$ *alone.*

Proof. Substituting $C_M^2 = C_P^2 \max(1, 4\varepsilon^2 m)$ from (4.38) into (4.59) in the form $\delta \lesssim \Delta t/(2C_M^2)$ gives (4.60). The maximum is attained by its first argument exactly when $4\varepsilon^2 m \le 1$; expanding m and using $\varepsilon \ll 1$ turns this into $\varepsilon\,\Delta t\,N \lesssim 1$, and the two branches then give (4.61) and (4.62), the latter after replacing m by its large-argument form $\Delta t^2 N^2/4$. Equating the two branches, $4\varepsilon^2 m = 1$ with $m \gg 1$, locates the cross-over at $\Delta t = 1/(\varepsilon N)$. ∎

The reversal of the Δt-scaling between the two regimes is the central finding of this section. The heuristic $\delta \propto \Delta t$, motivated by a change-of-variable argument on the related mixed Poisson problem [24], is the weakly stratified regime. Once the stratified branch of (4.38) takes over, the scaling inverts to $\delta \propto 1/\Delta t$. The competition is between a horizontal and a vertical mode, so the controlling group is $\varepsilon\,\Delta t\,N$ rather than $\Delta t\,N$. A thinner domain pushes the cross-over to larger time steps.

Since the numerical configurations of Section 5 are specified in physical units, we record the dimensional form of this condition, which is the form the experiments rely on.

COROLLARY 4.13 (Dimensional form of the admissible shift). *With the dimensional constants of Corollary 4.7, the admissible-shift condition $\delta \lesssim \Delta t/(2C_M^2)$ of (4.59) keeps the same form. In the two regimes of Proposition 4.12 it reads*

$$\delta \;\lesssim\; \frac{2\pi^2\,\Delta t}{L_x^2} \quad (\varepsilon\,\Delta t\,N \lesssim 1), \qquad \delta \;\lesssim\; \frac{2\pi^2}{\Delta t\,N^2 L_z^2} \quad (\varepsilon\,\Delta t\,N \gtrsim 1), \tag{4.63}$$

with the cross-over at $\Delta t \sim L_x/(NL_z) = 1/(\varepsilon N)$.

Proof. The shift itself is dimensional: comparing $\delta\langle\phi, p\rangle$ with $\langle\phi, \nabla\cdot\vec{u}\rangle$ in the shifted divergence equation (3.3a), with the kinematic pressure carrying $[p] = \mathrm{L}^2\,\mathrm{T}^{-2}$, gives $[\delta] = \mathrm{T}\,\mathrm{L}^{-2}$. Writing hats for the non-dimensional quantities of the preceding sections, with reference time T we have $\hat\delta = \delta L_x^2/T$ and $\widehat{\Delta t} = \Delta t/T$, so in the condition $\hat\delta \lesssim \widehat{\Delta t}/(2\hat C_M^2)$ of (4.59) the factors of T and L_x cancel, leaving $\delta \lesssim \Delta t/(2C_M^2)$ with the dimensional C_M of (4.39). Substituting the two branches of the maximum gives (4.63), and equating them locates the cross-over. ∎

In particular the conditions (4.63) are independent of the reference time used in the non-dimensionalisation, because the shift and the time step scale with it in the same way. Every practical threshold quoted in Section 5 is an instance of (4.63).

5. Numerical experiments. In this section we evaluate the preconditioner of Section 3 on two test cases.

The first, and the one used for all quantitative sweeps, is the Skamarock–Klemp gravity-wave test case [31], a flat-domain problem started from rest. It is used for three purposes: to demonstrate that the compatible finite element discretisation of Section 2.2 reproduces the gravity-wave dynamics of the linearised Boussinesq system (Section 5.1); to confirm that the shifted preconditioner is robust in the mesh parameters Δx, Δz, in the aspect ratio ε, and in the shift parameter, for both the linear system (Section 5.3) and the nonlinear problem (Section 5.4); and to compare the asymptotic scaling of δ established in Section 4 against the observed convergence behaviour.

The second is stratified flow over an isolated ridge (Section 5.5), which is a verification step. Its purpose is to check the solver against a problem that the first case does not exercise: a deformed, terrain-following mesh and a non-trivial mean flow. It also provides an independent test of the domain-size scaling of δ predicted in Section 4.5. Section 5.6 then turns from iteration counts to wall-clock cost, and Section 5.7 asks whether a smaller patch suffices.

In our implementation the two-dimensional slice mesh is embedded in a three-dimensional ambient space, so that all vector-valued finite element spaces remain three-dimensional and accommodate the three velocity components (u', v', w') in (2.13), despite the spatial dependence being only on x and z. MPI partitioning is performed on the one-dimensional base mesh, so that each MPI rank owns whole vertical columns; this preserves the column-wise data locality required by the solver of Section 3 and makes the column-patch smoother of Section 3.4 a purely local operation in the parallel implementation used for the timings of Section 5.6. The y-independence of the continuous solution is preserved by the symmetry of the equations and is inherited by the discrete spaces.

5.1. Test problem. We use the inertia–gravity wave test case of Skamarock and Klemp [31], in which a localised buoyancy perturbation in a stably stratified channel splits into two counter-propagating gravity-wave packets. The domain is $\Omega = (0, L_x) \times (0, L_z)$ with $L_x = 300\,\mathrm{km}$ and $L_z = 4\,\mathrm{km}$ for the nominal configuration (modified to $L_z \in \{12, 8, 4, 2, 1, 0.5, 0.25\}$ km for the aspect-ratio sweep). Buoyancy frequency is $N = 10^{-2}\ \mathrm{s}^{-1}$. Rotation is switched off, $f = 0$, so that the velocity block is symmetric and the configuration realises exactly the hypothesis of Theorem 4.10. Initial conditions are zero velocity and a localised buoyancy perturbation

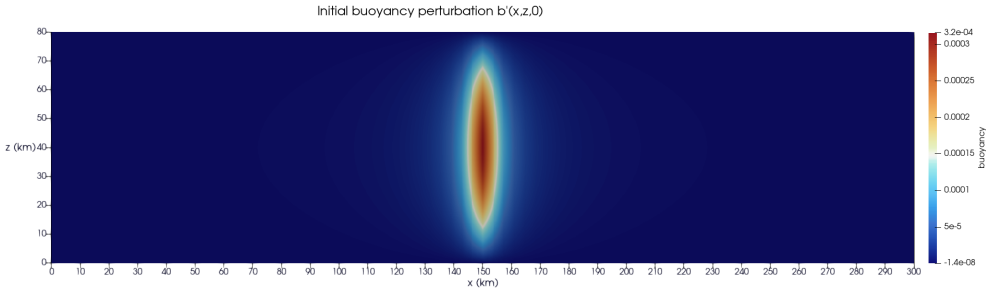
$$b_0'(x, z) = \Delta b_0 \, \frac{\sin(\pi z / L_z)}{1 + ((x - x_c)/a)^2},$$

with amplitude $\Delta b_0 = 0.01 (m\, s^{-2})$, centre $x_c = L_x/2$, and horizontal half-width $a = 5\,\mathrm{km}$. Boundary conditions are as in Section 2.1 periodic in the horizontal and no-flux on top and bottom.

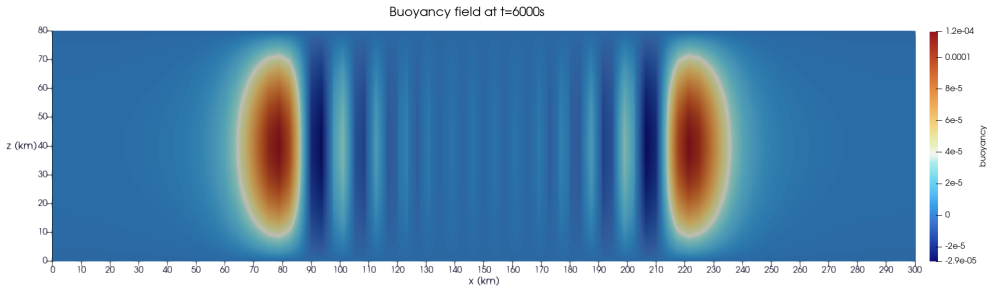
Figure 3 confirms that the compatible finite element discretisation of Section 2.2 reproduces the expected gravity-wave dynamics. The initially localised buoyancy perturbation (panel a) splits symmetrically into two outward-propagating wave packets visible at $t = 6000\,\mathrm{s}$ (panel b), with a phase speed consistent with the buoyancy frequency N.

5.2. Solver configuration. The compatible finite element triple of Section 2.2 is used with polynomial degree $k = 2$, giving $\mathrm{RT}_2 \otimes \mathbb{W}_h^\theta \otimes \mathrm{DG}_1$. Time integration uses the implicit midpoint rule through Irksome. fully discrete system at each time step is solved with FGMRES, right-preconditioned by the shifted Schur-complement preconditioner of Section 3. The convergence criterion is reduction of the solution error norm by ten orders of magnitude, where the solution error is measured against a direct-solver reference solution computed independently for each configuration. This diagnostic is a stronger indicator of solver quality than the GMRES residual, since it reflects the actual accuracy of the approximate solution rather than the residual norm at which the iteration terminates. The maximum iteration count is set to 100.

The shift parameter is parametrised as $\delta = C_1/\Delta t$ throughout, with nominal value $C_1 = 10^{-3}$; the sensitivity of the preconditioner to this choice of C_1 is examined in Section 5.3. We note at the outset that this is not the parametrisation the worst-case analysis of Section 4.5 would suggest for these parameters. With $\varepsilon = 1.3 \times 10^{-2}$ and



(a) Initial buoyancy perturbation $b'(x, z, 0)$.



(b) Buoyancy field at $t = 6000\,\mathrm{s}$.

FIG. 3. *Skamarock–Klemp test case on the domain* $\Omega = (0, 300\,\mathrm{km}) \times (0, 80\,\mathrm{km})$. *(a) The initial buoyancy perturbation* $b'(x, z, 0)$*, localised near the horizontal centre* $x_c = 150\,\mathrm{km}$ *and concentrated in the vertical by the* $\sin(\pi z/L_z)$ *envelope. (b) Buoyancy field at* $t = 6000\,\mathrm{s}$*: the initial perturbation has split into two symmetric wave packets propagating outward in opposite horizontal directions, consistent with the internal gravity-wave phase speed determined by the buoyancy frequency* N*.*

$N = 10^{-2}\,\mathrm{s}^{-1}$ the configuration lies in the weakly stratified regime of Proposition 4.12, where the admissible shift grows as $\delta \propto \Delta t$, whereas $\delta = C_1/\Delta t$ decreases with Δt. Holding C_1 fixed therefore moves the solver further inside the admissible range as Δt grows, at a rate Δt^2, and this is what the sweeps of Section 5.3 exhibit at small Δt.

The inner solvers are configured as in Section 3.4: an exact block-diagonal solve on the pressure mass matrix, and a multigrid V-cycle on the velocity-buoyancy block with column-patch additive Schwarz smoothing via Firedrake's `ASMStarPC` (`pc_star_construct_dim:0`). Reverse Cuthill–McKee reordering is applied to the patch subproblems prior to the local LU factorisations for better ordering of the sparse matrix.

5.3. Parameter robustness. We assess robustness of the preconditioner along four axes: the horizontal mesh resolution Δx, the vertical mesh resolution Δz, the aspect ratio $\varepsilon = L_z/L_x$, and the shift parameter scaling C_1. In each case the remaining parameters are held at their nominal values, and the time step is swept over $\Delta t \in \{1, 5, 10, 50, 100, 500, 10^3, 5 \times 10^3, 10^4\}$ s. The metric reported is the number of GMRES iterations required to achieve the solution-error tolerance.

Figure 4 shows the FGMRES convergence history at $\Delta t = 50\,\mathrm{s}$ for five horizontal resolutions, $N_x \in \{10, 20, 40, 60, 80\}$. The curves nearly collapse, indicating that the iteration count is essentially independent of Δx across different horizontal resolution.

The analogous study for the vertical resolution is shown in Figure 5, with $N_z \in \{10, 20, 40, 60, 80\}$. The iteration counts are even more tightly clustered than in the horizontal sweep, reflecting the fact that the column-patch smoother resolves the vertical coupling exactly within each patch.

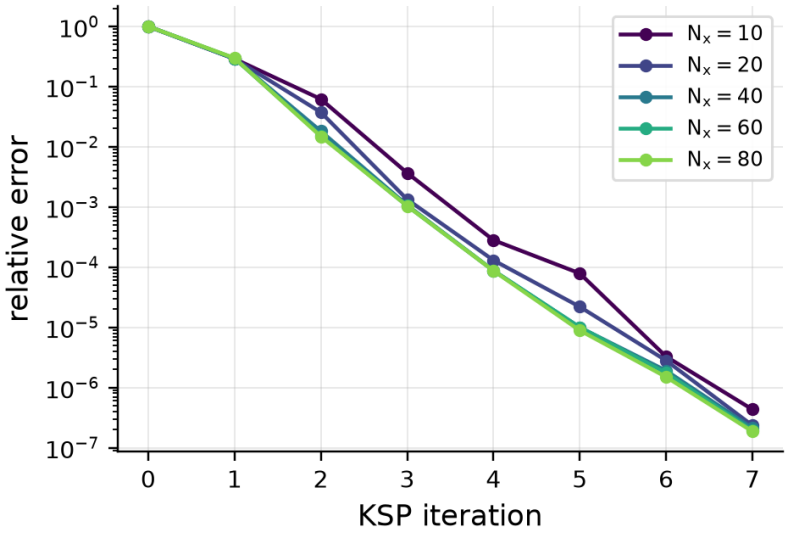


FIG. 4. *Robustness in the horizontal resolution Δx. FGMRES convergence history (relative solution-error norm vs iteration index) for $N_x \in \{10, 20, 40, 60, 80\}$ at fixed $\Delta t = 50\,\mathrm{s}$, with remaining parameters held at their nominal values.*

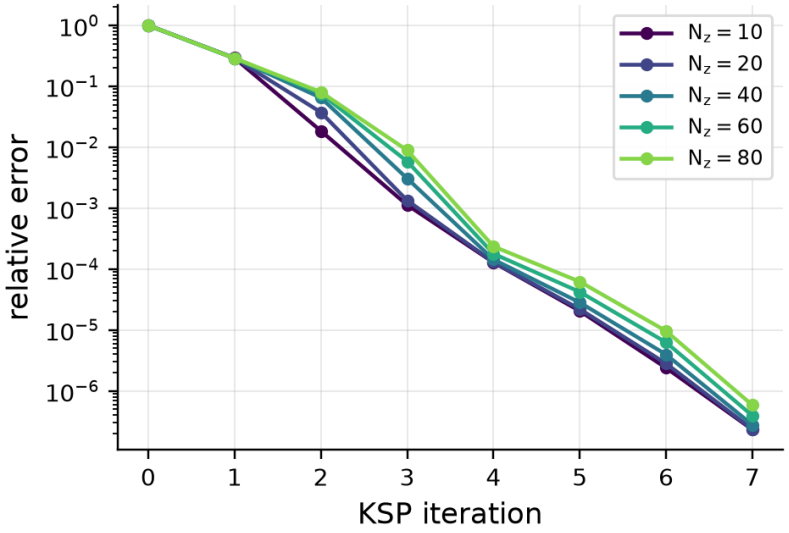


FIG. 5. *Robustness in the vertical resolution Δz. FGMRES convergence history for $N_z \in \{10, 20, 40, 60, 80\}$ at fixed $\Delta t = 50\,\mathrm{s}$, with remaining parameters held at their nominal values. The curves are more tightly clustered than in Figure 4, reflecting the fact that the column-patch smoother resolves the vertical coupling exactly within each patch.*

Figure 6 sweeps the aspect ratio over $\varepsilon \in \{4\times10^{-2}, 2.67\times10^{-2}, 1.33\times10^{-2}, 6.67\times10^{-3}, 3.33\times10^{-3}, 1.67\times10^{-3}, 8.3\times10^{-4}\}$, spanning two orders of magnitude, by varying L_z at fixed L_x. Thinner domains never converge more slowly, in agreement with Corollary 4.6: for $\varepsilon\,\Delta t\,N \lesssim 1$, which holds throughout this sweep, the inf–sup constant β is exactly independent of ε, and so is every other constant entering the bound of Theorem 4.11. The figure shows more than this: the iteration count falls

monotonically as the domain thins, from 18 at the largest aspect ratio to 3 at the smallest. A bound that is uniform in ε permits this but does not explain it; the improvement comes from the vertical branch of Lemma 4.5, the constant seen by data with vertical structure, which improves like ε^2 as the domain thins. Remark 5.1 sets this out, and it is worth doing because the same mode structure also accounts for the quantitative gap between the analysis and the shift sweep below.

Remark 5.1 (Why thinner domains converge faster). With $f = 0$ the pressure can be eliminated exactly. Writing $A : V \to V'$ and $B : V \to Q'$ for the operators induced by the forms a and b, and $S := BA^{-1}B^*$ for the pressure Schur complement of the unshifted problem, the first equation of (4.5) gives $\vec{u} = -A^{-1}B^*p$, and the second then reduces to an equation for the pressure component of $(\mathcal{A}_\delta^{-1}\mathcal{A}_0 - I)(\vec{u}_0, p_0)$ alone,

$$(S + \delta I)\, p = -\delta\, p_0, \qquad \text{so that} \qquad p = -\delta\, (S + \delta I)^{-1} p_0. \tag{5.1}$$

On the box S is diagonal in the separable basis ψ_{jl} of Lemma 4.5, with eigenvalues

$$\sigma_{jl} = \Delta t \left(A_j + \frac{B_l}{m} \right) = \Delta t \left[\left(\frac{2\pi j}{L_x} \right)^2 + \frac{1}{m} \left(\frac{\pi l}{L_z} \right)^2 \right] \tag{5.2}$$

in physical units, because $\langle S\psi, \psi \rangle = \sup_{\vec{v}} b(\vec{v}, \psi)^2 / a(\vec{v}, \vec{v})$ is Δt times the eigenvalue of $-\nabla \cdot M^{-1} \nabla$ on ψ. Each pressure mode is therefore damped by its own factor

$$\rho_{jl} = \frac{\delta}{\delta + \sigma_{jl}}, \tag{5.3}$$

and the constants of Section 4 are built from the worst of these: $\min_{j,l} \sigma_{jl} = \Delta t / C_M^2$, the two branches of $C_M^2 = \max(C_h^2, m C_v^2)$ being the modes $(1, 0)$ and $(0, 1)$. In the thin-domain regime the minimum is attained by the depth-independent mode $(1, 0)$, whose $\sigma_{10} = 4\pi^2 \Delta t / L_x^2$ does not involve L_z at all; this is the mode that makes the bound independent of ε. Every mode with vertical structure, $l \geq 1$, has instead $\sigma_{jl} \geq \sigma_{0l} = \pi^2 l^2 \Delta t / (m L_z^2)$, the vertical branch of Lemma 4.5, which grows like L_z^{-2} as the domain thins.

The test problem of Section 5.1 excites only modes of the second kind. Its buoyancy perturbation is proportional to $\sin(\pi z / L_z)$ and vanishes on both lids, so the depth average of the pressure forcing $\partial_z b$ is zero; and since N is constant the linearised problem separates in z, so no depth-independent component is generated at any later time or in any Krylov iterate. The solver therefore only ever sees modes whose damping factors lie on the vertical branch. Table 1 evaluates the largest of them, ρ_{01}, together with ρ_{10}, for the seven aspect ratios of Figure 6 at $\Delta t = 50\,\mathrm{s}$ and $\delta = C_1 / \Delta t$ with $C_1 = 10^{-3}$. The modes present go from barely damped at $\varepsilon = 4 \times 10^{-2}$ to essentially eliminated at $\varepsilon = 8.3 \times 10^{-4}$, which is the ordering, and roughly the spacing, of the curves in Figure 6; the depth-independent mode, on which the bound of Section 4 rests, would be damped by 0.999 at every aspect ratio, exactly as that bound predicts. The bound is flat in ε because it is sharp for a mode that the data do not contain; the improvement with ε is the vertical branch of Lemma 4.5 acting on the modes they do.

We finally study the dependence of the iteration count on the shift parameter scaling C_1 in $\delta = C_1 / \Delta t$. Figure 7 shows the iteration count as a function of Δt for six values of C_1 spanning five orders of magnitude, $C_1 \in \{1, 10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$. The curves separate into two regimes corresponding to the asymptotic analysis of Section 4.

TABLE 1

Mode damping factors (5.3) for the aspect-ratio sweep of Figure 6, at $\Delta t = 50\,\mathrm{s}$, $N = 10^{-2}\,\mathrm{s}^{-1}$ and $C_1 = 10^{-3}$. The last column is the iteration count at which the corresponding curve of Figure 6 terminates.

L_z (km)	ε	ρ_{01} (vertical branch)	ρ_{10} (horizontal branch)	Iterations
12	4.0×10^{-2}	0.86	0.999	18
8	2.7×10^{-2}	0.73	0.999	13
4	1.3×10^{-2}	0.41	0.999	7
2	6.7×10^{-3}	0.15	0.999	5
1	3.3×10^{-3}	0.041	0.999	4
0.5	1.7×10^{-3}	0.011	0.999	3
0.25	8.3×10^{-4}	0.0027	0.999	3

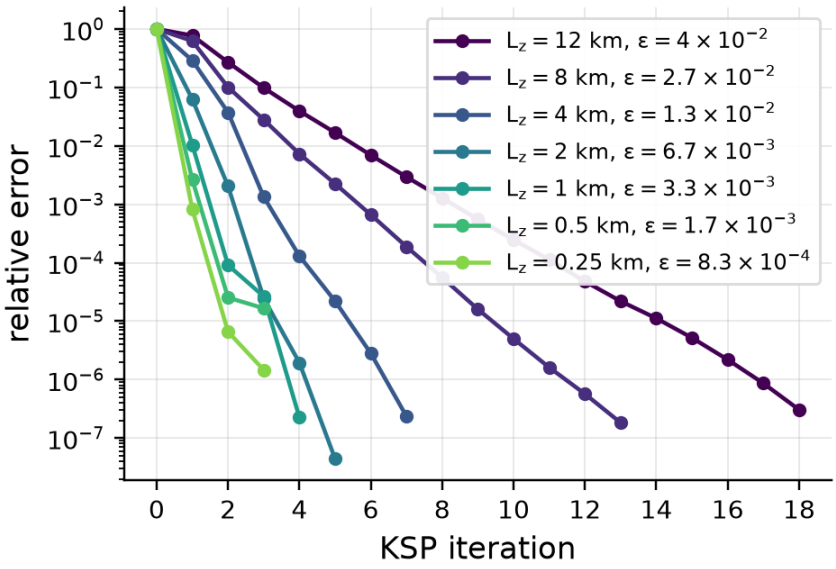


FIG. 6. *Robustness in the aspect ratio $\varepsilon = L_z/L_x$. FGMRES convergence history for seven aspect ratios at fixed $\Delta t = 50\,\mathrm{s}$. Thinner domains (smaller ε) converge in fewer iterations. The bound of Theorem 4.11 is uniform in ε; the improvement is explained in Remark 5.1.*

With $\varepsilon \approx 1.3 \times 10^{-2}$ and $N = 10^{-2}\,\mathrm{s}^{-1}$, the cross-over of Proposition 4.12 lies at $\Delta t \sim 1/(\varepsilon N) \approx 7.5 \times 10^3\,\mathrm{s}$, beyond the range of time steps tested, so the nominal configuration lies inside the weakly stratified regime, and rotation is switched off; the applicable result is therefore the symmetric Theorem 4.10 and the prediction is the single condition $\delta \lesssim \Delta t/(2C_P^2)$ with $C_P = L_x/(2\pi)$, the dimensional form given in Corollary 4.13. In terms of the parametrisation $\delta = C_1/\Delta t$ this reads

$$C_1 \lesssim \frac{2\pi^2\,\Delta t^2}{L_x^2} \approx 2 \times 10^{-10}\,\Delta t^2, \tag{5.4}$$

a threshold that tightens quadratically as Δt decreases. Two qualitative consequences follow, and both are visible in Figure 7: at fixed Δt the preconditioner degrades once C_1 exceeds the threshold, and at fixed C_1 it improves as Δt grows. For $C_1 \in \{1, 10^{-1}\}$ the iteration count diverges (the $C_1 = 1$ curve never converges to the tolerance within

the iteration cap for $\Delta t \le 10^2$), while for $C_1 \le 10^{-2}$ it remains bounded and decreases as Δt grows; and at $\Delta t = 1$ the iteration count is best at $C_1 = 10^{-5}$, the smallest value tested, as the quadratic tightening requires.

Quantitatively, however, (5.4) is far stricter than the data require. At $\Delta t = 10^2\,\mathrm{s}$ it gives $C_1 \lesssim 2 \times 10^{-6}$, whereas the solver converges comfortably at $C_1 = 10^{-3}$ and the observed boundary between bounded and diverging iteration counts lies near $C_1 = 10^{-2}$, three to four orders of magnitude higher. Remark 5.1 explains the gap. The condition (5.4) is the leading-order condition $2\delta/\sigma < 1$ of Section 4.5 evaluated on the depth-independent pressure mode, $\sigma = \sigma_{10}$ in (5.2); it is sharp for that mode, but the test data do not contain it. The modes they do contain lie on the vertical branch, and the same condition evaluated on the least damped of them, $\sigma = \sigma_{01}$, reads

$$C_1 \lesssim \frac{\pi^2\,\Delta t^2}{2\,m\,L_z^2} = \frac{\pi^2\,\Delta t^2}{2L_z^2\,(1 + \Delta t^2 N^2/4)}, \tag{5.5}$$

which is 7×10^{-4} at $\Delta t = 50\,\mathrm{s}$, 2.5×10^{-3} at $10^2\,\mathrm{s}$, 10^{-2} at $5 \times 10^2\,\mathrm{s}$, and saturates at $2\pi^2/(N^2 L_z^2) \approx 1.2 \times 10^{-2}$ for $\Delta t \gg 2/N$. This reproduces the observed boundary near 10^{-2} for $\Delta t \ge 10^2\,\mathrm{s}$ to within a factor of a few, and at $\Delta t \le 5\,\mathrm{s}$, where no tested C_1 lies below it, the counts are large for every C_1 and ordered monotonically in it. It also explains the shape of the curves in Figure 7: for $\Delta t \ll 2/N = 200\,\mathrm{s}$ the threshold tightens as Δt^2, so at fixed C_1 the iteration count falls steeply as Δt grows, whereas for $\Delta t \gg 2/N$ the threshold is independent of Δt and the counts level off. The saturated value is precisely the strongly stratified branch of Corollary 4.13, $\delta \lesssim 2\pi^2/(\Delta t\, N^2 L_z^2)$, which for these data is entered at $\Delta t\, N \sim 2$ rather than at $\varepsilon\,\Delta t\, N \sim 1$: the cross-over of Proposition 4.12 at $1/(\varepsilon N) \approx 7.5 \times 10^3\,\mathrm{s}$ marks where the vertical branch of C_M overtakes the horizontal one, and is visible only to data that contain the horizontal mode, whereas on the vertical branch alone the change from $\delta \propto \Delta t$ to $\delta \propto 1/\Delta t$ happens as soon as m departs from one. What the worst-case analysis predicts correctly for every right-hand side is thus the quadratic tightening at small Δt and the direction of both trends; (5.4) is a guarantee, and (5.5) is its sharp form for right-hand sides without a depth-independent pressure component. [TODO: A sweep at $\Delta t \sim 1/(\varepsilon N) \approx 7.5 \times 10^3$ s with initial data that contain a depth-independent component, such as a barotropic velocity or a buoyancy profile that does not vanish at the lids, would exhibit the worst-case cross-over of Proposition 4.12 and test (5.4) directly; the present data cannot reach either.]

The four sweeps together establish that the preconditioner is robust in Δx, Δz, ε, and (within the analytically predicted range of) C_1. Once the mode content of the data is taken into account as in Remark 5.1, the agreement with the analysis of Section 4 is quantitative: (5.5) locates the convergence boundary to within a factor of a few, the levelling-off at $\Delta t \sim 2/N$ is the predicted change of scaling, and the improvement with decreasing ε is the vertical branch of Lemma 4.5. This supports the use of the analytical scaling as a practical guide for selecting the shift parameter, with the guarantee (5.4) reserved for right-hand sides that may contain a depth-independent pressure component.

5.4. Extension to the nonlinear problem. The shifted Schur-complement preconditioner is designed for the linearised system arising at each Newton step of a nonlinear solve, and we now verify that this coupling is effective in practice. The full nonlinear Boussinesq equations are discretised with the same compatible finite element triple and time integrator as in Section 5.2, the outer nonlinear iteration is handled by

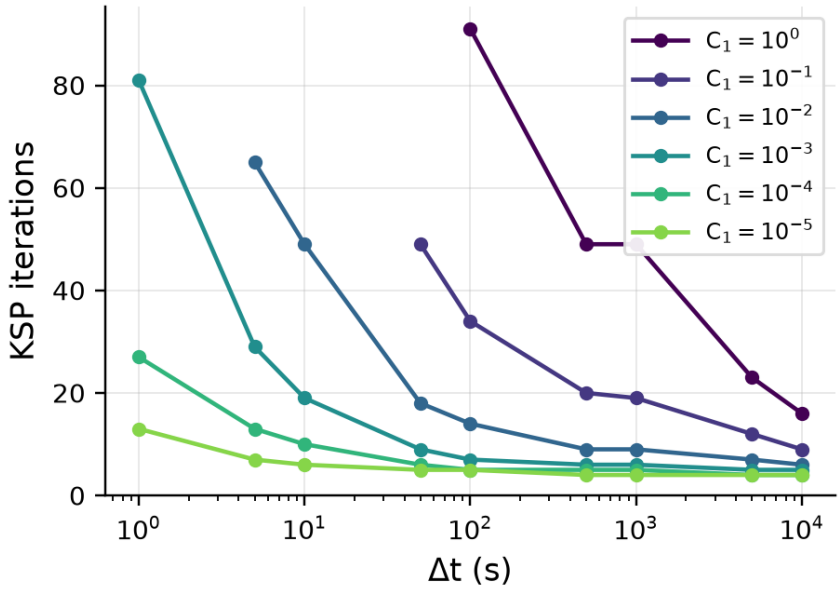


FIG. 7. *Robustness in the shift parameter scaling. FGMRES iteration count as a function of* Δt *for* $C_1 \in \{10^0, 10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$ *in* $\delta = C_1/\Delta t$*, with remaining parameters held at their nominal values. The worst-case threshold* (5.4) *applies to every right-hand side; for the data used here the operative threshold is* (5.5)*, which tightens as* Δt^2 *for* $\Delta t \ll 2/N$ *and is constant, near* 10^{-2}*, for* $\Delta t \gg 2/N$*. The curves separate accordingly into those that respect it and those that do not, and level off beyond* $\Delta t \sim 2/N$*.*

PETSc's SNES (Newton–Krylov), and the inner linear system at each Newton step is solved with FGMRES preconditioned by A_δ^{-1} as before. The convergence metric is the relative solution error $\|U_k - U^*\|/\|U^*\|$ plotted against the *cumulative* number of inner KSP iterations across all Newton steps; each change of slope in the curve corresponds to the start of a new Newton step.

Figures 8 and 9 confirm that the preconditioner transfers to the nonlinear regime without degradation: the curves for different horizontal and vertical resolutions are tightly clustered, matching the resolution independence observed in the linear experiments of Section 5.3.

Figure 10 repeats the aspect-ratio sweep of Section 5.3 for the nonlinear problem at two representative time steps, $\Delta t = 100\,\mathrm{s}$ and $\Delta t = 5000\,\mathrm{s}$. In both cases the curves are tightly clustered, confirming that the preconditioner retains its aspect-ratio robustness in the nonlinear setting.

Figures 11 and 12 examine the sensitivity of the nonlinear solver to the shift parameter. Figure 11 shows the convergence history at $\Delta t = 500\,\mathrm{s}$ for seven values of C_1; Figure 12 shows the total KSP iteration count as a function of Δt across the full time-step range. The qualitative picture mirrors the linear case of Section 5.3: a well-chosen C_1 in the range $[10^{-4}, 10^{-3}]$ gives uniformly low iteration counts, and the change of scaling occurs near $\Delta t \approx 100\,\mathrm{s}$, which is the cross-over $\Delta t\, N \sim 1$ expected for data without an appreciable depth-independent pressure component (Remark 5.1), not the worst-case cross-over at $1/(\varepsilon N)$.

5.5. A second nonlinear test case: stratified flow over a ridge. The Skamarock–Klemp case of Section 5.1 is posed on a flat, undeformed domain with no

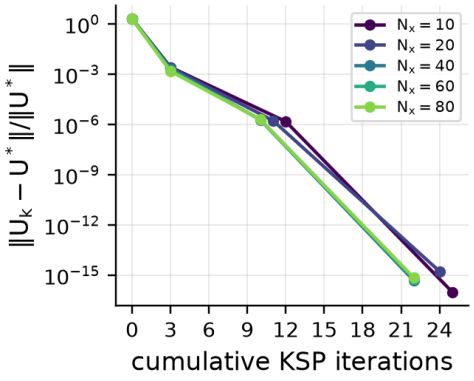


FIG. 8. *Nonlinear robustness in Δx: relative solution error vs cumulative inner KSP iterations for five horizontal resolutions at $\Delta t = 50\,\mathrm{s}$.*

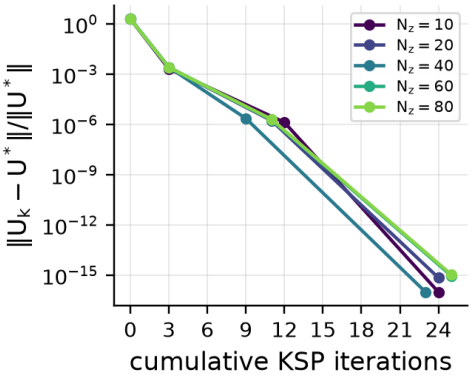
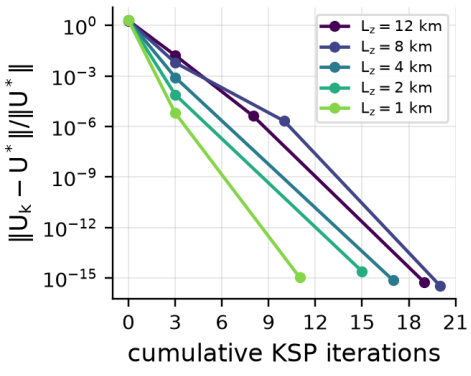


FIG. 9. *Nonlinear robustness in Δz: relative solution error vs cumulative inner KSP iterations for five vertical resolutions at $\Delta t = 50\,\mathrm{s}$.*



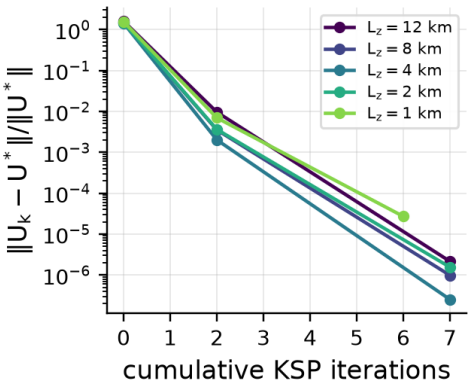


FIG. 10. *Nonlinear robustness in the aspect ratio $\varepsilon = L_z/L_x$: relative solution error vs cumulative inner KSP iterations at $\Delta t = 100\,\mathrm{s}$ (left) and $\Delta t = 5000\,\mathrm{s}$ (right) for five aspect ratios. The curves are tightly clustered at both time steps.*

mean flow. To check that the preconditioner survives contact with a problem that is not of that form, we ran a second nonlinear test case: stratified flow over an isolated ridge, following the vertical-slice configuration of Cotter and Shipton [8, Section 3.3] and originally due to Melvin et al. [28]. It exercises three features the first case does not: a *deformed, terrain-following mesh*, on which the vertical column structure that the star smoother of Section 3.4 relies on must be preserved by the mesh hierarchy; a *non-trivial mean flow*, which activates the advective terms of the nonlinear residual; and a *sponge layer* near the model top. This section reports verification that the discretisation reproduces the expected solution and that the solver converges at the expected rate. We have not repeated the parameter sweeps of Section 5.3 on this configuration.

We follow the configuration of Melvin et al. [28], in the compatible finite element formulation of Cotter and Shipton [8, Section 3.3]. The domain is a slice of length L and height H, periodic in x, with a rigid lid and a lower boundary raised into a

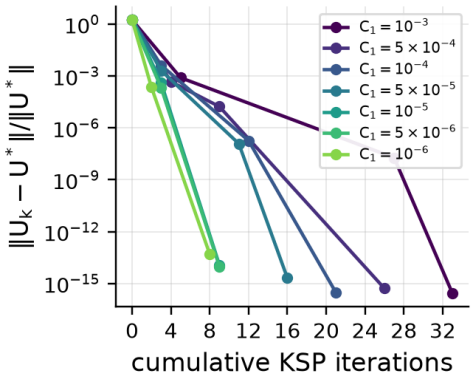


FIG. 11. *Nonlinear shift-parameter sensitivity: convergence history at* $\Delta t = 500\,\mathrm{s}$ *for* $C_1 \in [10^{-3}, 10^{-6}]$.

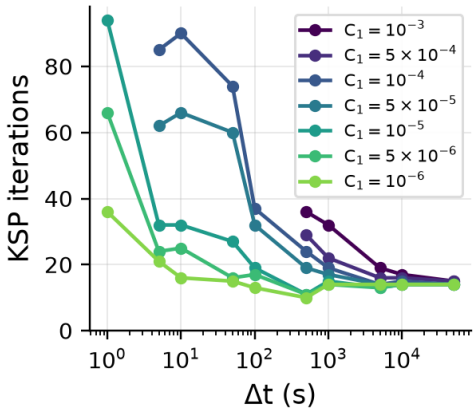


FIG. 12. *Nonlinear shift-parameter sweep: total KSP iteration count vs* Δt *for* $C_1 \in [10^{-3}, 10^{-6}]$.

"witch of Agnesi" ridge

$$z_s(x) = \frac{h_m\, a^2}{x^2 + a^2}, \tag{5.6}$$

imposed through the terrain-following coordinate map

$$(x, z) \longmapsto \left(x,\; z + z_s(x)\, \frac{H - z}{H} \right), \tag{5.7}$$

which deforms rectangular cells into trapezia while leaving the mesh columns vertical, the deformation decaying linearly to zero at the model top. A uniform horizontal wind $u = (U, 0, 0)$ is switched on impulsively over a stably stratified fluid at rest. The only forcing is the no-normal-flow condition on the sloping ground, which launches a stationary gravity-wave train above the ridge. A Newtonian sponge damps the vertical velocity near the lid so that upward-propagating waves are absorbed rather than reflected,

$$\mu(z) = \begin{cases} 0, & z < z_B, \\ \bar{\mu} \sin^2\!\left(\dfrac{\pi}{2}\, \dfrac{z - z_B}{H - z_B} \right), & z \geq z_B, \end{cases} \tag{5.8}$$

entering the momentum equation as $\mu(z)\,(\vec{u}\cdot\hat{k})\,\hat{k}$, so that the buoyancy is not damped. Parameters are collected in Table 2.

In both configurations $Nh_m/U \leq 10^{-3} \ll 1$, so the flow remains in the linear regime in which the analytic Queney solution applies and the Boussinesq and compressible answers should agree; the expected vertical velocity scale is $w \sim Uh_m/a$. The aspect ratio, $\varepsilon = H/L \approx 0.2$ in both cases, is an order of magnitude larger than the nominal value in the Skamarock–Klemp sweeps of Section 5.3 and larger still than the thinnest cases there, so this test probes a different corner of the parameter space rather than repeating one already covered; it nonetheless remains inside the range $\varepsilon \leq \frac{1}{2}$ assumed in Corollary 4.6. The two regimes are distinguished by the nonhydrostatic parameter Na/U, reported in the last row of Table 2: the hydrostatic

TABLE 2
Parameters for the ridge test case, after Melvin et al. [28] as configured in [8, Section 3.3]. The two regimes differ in the domain, the mean wind and the time step as well as in the ridge half-width.

Quantity	Symbol	Nonhydrostatic	Hydrostatic
Domain length	L	144 km	240 km
Domain height	H	35 km	50 km
Columns × layers	–	180 × 70	100 × 60
Ridge half-width	a	1 km	10 km
Ridge height	h_m	1 m	1 m
Mean wind	U	10 m s^{-1}	20 m s^{-1}
Buoyancy frequency	N	10^{-2} s^{-1}	10^{-2} s^{-1}
Time step	Δt	5 s	20 s
Final time	T	9000 s	15000 s
Sponge base	z_B	25 km	30 km
Sponge strength	$\bar{\mu}\,\Delta t$	0.15	0.3
Rotation	f	0	0
Aspect ratio	$\varepsilon = H/L$	0.24	0.21
Nonhydrostatic parameter	Na/U	1	5

approximation is the wide-ridge, large-Na/U limit, and the two configurations sit at $Na/U = 1$ and $Na/U = 5$ respectively.

One implementation point concerning the initial condition is worth recording, because it is specific to the combination of a deformed mesh with a stage-value formulation of the time integrator. Irksome imposes both the algebraic constraint $\nabla\cdot\vec{u} = 0$ and the Dirichlet data on the Runge–Kutta *stage values* rather than on $\vec{u}^{\,n+1}$ directly. Under the implicit midpoint rule the constrained quantity is $(\vec{u}^{\,n} + \vec{u}^{\,n+1})/2$, so that

$$\nabla\cdot\vec{u}^{\,n+1} = -\nabla\cdot\vec{u}^{\,n}, \qquad \vec{u}^{\,n+1}\cdot\hat{n}\big|_{\partial\Omega} = -\,\vec{u}^{\,n}\cdot\hat{n}\big|_{\partial\Omega}. \tag{5.9}$$

An initial state that is not solenoidal, or that does not respect no-normal-flow on the slope, is therefore never damped out. It simply changes sign at every step. On a flat mesh this is invisible, since the Raviart–Thomas projection of a constant vector is exactly divergence free, but on the deformed mesh it is not. The velocity is accordingly initialised by a discrete Leray projection: find $(\vec{u}, p) \in H_h(\mathrm{div}) \times L^2_h$ with $\vec{u}\cdot\hat{n} = 0$ on the top and bottom such that

$$\langle\vec{w},\vec{u}\rangle - \langle\nabla\cdot\vec{w},\, p\rangle = \langle\vec{w},\,(U,0,0)\rangle, \qquad \langle\phi,\,\nabla\cdot\vec{u}\rangle = 0, \tag{5.10}$$

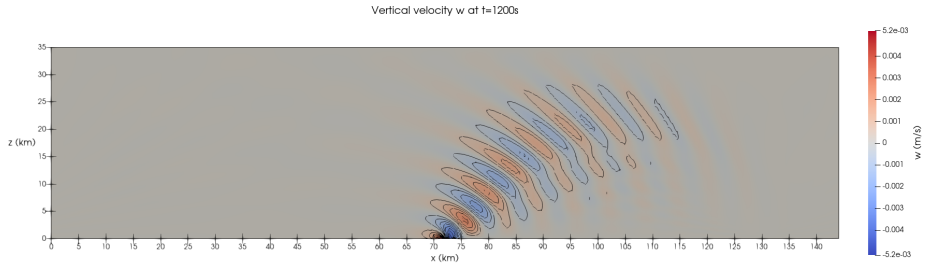
for all $(\vec{w}, \phi)$.

The shift was set by holding the dimensionless group $\delta\,UL$ fixed at 10^{-4} across both configurations of Table 2, that is $\delta = 6.9\times10^{-11}$ s m^{-2} in the nonhydrostatic case; carrying the unit-box value $\delta = 10^{-4}$ s m^{-2} over unchanged stalls the Krylov solver, so the domain-size dependence of the admissible shift (Corollary 4.13) is essential here. The two test cases are also consistent with each other once the mode content of the data is taken into account: on the Skamarock–Klemp configuration the solver operates above the worst-case threshold and still converges, for the reason given in Remark 5.1, whereas on the ridge configuration the working value sits below that threshold and the unit-box value that fails sits far above it. The figures quoted below are for the nonhydrostatic configuration.

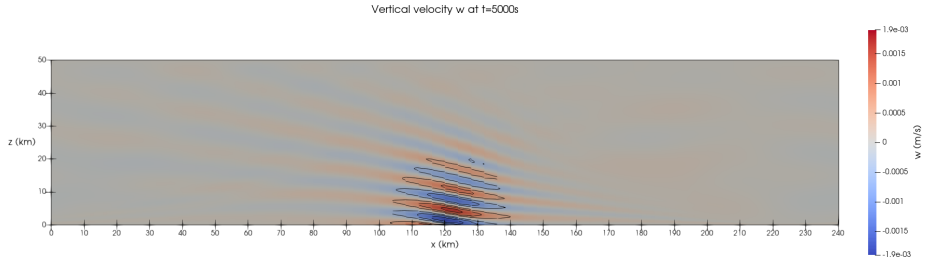
TABLE 3
Verification of the ridge test case at 180×70 *resolution with* $\Delta t = 5\,\mathrm{s}$.

Check	Result
$\|\nabla \cdot \vec{u}_0\|_{L^2}$ after projection	1.5×10^{-12}
Normal flow through the terrain, $\int_{\Gamma_b} \|\vec{u}_0 \cdot \hat{n}\|$	2.9×10^{-15}
Newton convergence, first step	$1.31 \to 4.8 \times 10^{-6} \to 3.6 \times 10^{-12}$
Linear solves per Newton step	4–7 iterations to `rtol`
KSP iterations per time step	11.7
$\max \|w\|$	$5.6 \times 10^{-3}\,\mathrm{m\,s^{-1}}$, steady

Table 3 collects the verification checks, run at the full resolution of Table 2. The initial projection delivers a discretely divergence-free field respecting the terrain boundary condition to round-off, the Newton and Krylov iteration counts are those expected from Section 5.4, the peak vertical velocity matches the linear estimate $Uh_m/a = 10^{-2}\,\mathrm{m\,s^{-1}}$ and the contour range reported in [28], and a flat-terrain control run ($h_m = 0$) gives an initial nonlinear residual of 1.2×10^{-9} against 1.31 with the ridge in place, so the background state is preserved to nine digits and the mountain signal sits nine orders of magnitude above the numerical floor. The solver runs the full 240 time steps without loss of robustness, and the wave field remains steady once established. Figure 13 shows the vertical velocity after 240 and 250 time steps ($t = 1200\,\mathrm{s}$ and $5000\,\mathrm{s}$) in the two regimes, while the wave train is still developing: in the nonhydrostatic case it propagates upward and downstream of the ridge, while in the hydrostatic case it remains confined above the ridge, as in the reference solutions.



(a) Nonhydrostatic regime ($a = 1\,\mathrm{km}$, $Na/U = 1$), after 240 time steps, $t = 1200\,\mathrm{s}$.



(b) Hydrostatic regime ($a = 10\,\mathrm{km}$, $Na/U = 5$), after 250 time steps, $t = 5000\,\mathrm{s}$.

FIG. 13. *Contours of the vertical velocity w for stratified flow over the witch-of-Agnesi ridge in the two regimes of Table 2, with contour lines every $5 \times 10^{-4}\,\mathrm{m\,s^{-1}}$ as in [8, Figs. 4–5]. The frames are taken before the final times of that table, so the wave train is still developing, but it already shows the character of the reference solutions: propagating upward and downstream of the ridge in the nonhydrostatic regime, and confined above the ridge in the hydrostatic regime.*

TABLE 4
Effect of RCM reordering on the column-patch LU factorisations. Aspect ratio $\varepsilon \sim 10^{-3}$, $\Delta t = 100\,\mathrm{s}$, $\delta = 10^{-4}$.

	RCM reordering			No RCM reordering		
(N_x, N_z)	Bandwidth	Time(PCApply)	Time/N_z	Bandwidth	Time(PCApply)	Time/N_z
$(20, 10)$	48	1.685s	0.169s	531	3.005s	0.300s
$(20, 20)$	48	2.628s	0.131s	1051	9.495s	0.475s
$(20, 40)$	48	4.630s	0.116s	1571	35.01s	0.875s
$(20, 80)$	48	8.005s	0.100s	2091	139.7s	1.746s

5.6. Timing. We now turn from iteration counts to the wall-clock cost of applying the preconditioner, examining in turn the effect of reordering the patch subproblems and the cost of the Schur-complement reduction relative to a monolithic alternative. Unless stated otherwise, the timings below are obtained on 16 MPI processes on a workstation with 64 GB of memory, using the two-dimensional extruded vertical-slice mesh of Section 2.2.

Each column-patch subproblem (3.29) is solved by a direct LU factorisation, and the cost of that factorisation depends strongly on the bandwidth of the patch matrix, which is determined by the ordering of the degrees of freedom within the patch. We apply reverse Cuthill–McKee (RCM) reordering [9, 18] to each patch prior to factorisation: RCM relabels the unknowns by breadth-first traversal of the patch sparsity graph, reversing the resulting Cuthill–McKee order, which is known to substantially reduce the matrix bandwidth (and hence the fill-in and factorisation cost) for the kind of locally-connected, line-like sparsity pattern that a column patch produces. Because the sparsity pattern of a column patch depends only on the local mesh topology and not on the patch data, the reordering is computed once per patch and reused across all FGMRES iterations and all time steps.

Table 4 reports the patch matrix bandwidth and the wall-clock cost of a single preconditioner application (PCApply), with and without RCM reordering, for a fixed aspect ratio $\varepsilon \sim 10^{-3}$, $\Delta t = 100\,\mathrm{s}$, and $\delta = 10^{-4}$, as the number of vertical layers N_z is increased at fixed $N_x = 20$. Without reordering the bandwidth grows with N_z and the resulting PCApply cost grows worse than linearly, reaching 139.7 s at $N_z = 80$; with RCM reordering the bandwidth is independent of N_z and the PCApply cost per layer, reported in the right-most column of each block, is essentially constant, consistent with a linear-in-N_z scaling of the total cost (a least-squares fit of Time(PCApply) against N_z for the RCM-reordered solves gives $R^2 = 0.998$). RCM reordering is therefore essential to realising the column-patch smoother of Section 3.4 as a practical line solve rather than a dense factorisation whose cost grows with the column length.

We next compare the fieldsplit Schur-complement approach of Section 3.3, in which pressure is eliminated exactly and the column-patch smoother of Section 3.4 acts only on the velocity-buoyancy block S, against a monolithic alternative in which the column-patch smoother is instead applied directly to the full saddle-point system A_δ, with velocity, buoyancy, and pressure degrees of freedom all coupled within each patch. This monolithic construction is the column-patch instance of the approach of Farrell, Mitchell, and Wechsung [16] discussed in Section 3.4.

Table 5 reports, for the same parameter values as Table 4, the number of degrees of freedom per patch, the cumulative wall-clock cost spent in preconditioner

TABLE 5
Fieldsplit Schur-complement preconditioning versus a monolithic patch solver applied directly to A_δ. Aspect ratio $\varepsilon \sim 10^{-3}$, $\Delta t = 100\,\mathrm{s}$, $\delta = 10^{-4}$, 16 MPI processes.

	Fieldsplit Schur			Monolithic		
(N_x, N_z)	Patch DOFs	Time(PCApply)	Its	Patch DOFs	Time(PCApply)	Its
$(10, 10)$	308	1.11s	12	388	0.63s	8
$(20, 20)$	608	2.61s	13	768	1.91s	7
$(40, 40)$	1208	8.01s	12	1528	6.33s	7
$(40, 80)$	2408	14.70s	12	3408	11.69s	7

application (PCApply) over the full FGMRES solve, and the resulting iteration count to the solution-error tolerance of Section 5.2, for both approaches across four mesh resolutions. Because the monolithic patches additionally couple the pressure degrees of freedom on each column, they are at least 20% larger than the fieldsplit Schur patches. This is reflected in the per-iteration application cost, Time(PCApply)/Its, which is lower for the fieldsplit Schur approach at three of the four resolutions tested (for instance 0.67 s versus 0.90 s at $(40, 40)$); the exception is the coarsest mesh, where per-patch overheads not reflected in the DOF count dominate. Solving each patch in the full saddle-point variables, however, resolves more of the pressure-velocity coupling per relaxation sweep than the Schur reduction, which defers that coupling to the grad-div augmentation of (3.12); the monolithic approach therefore converges in consistently fewer FGMRES iterations, and this advantage in iteration count outweighs its larger per-iteration cost, so that the cumulative PCApply time in Table 5 is in fact lower for the monolithic approach at every resolution tested in the present, untuned configuration. We do not attempt to tune the relative damping factors or smoother sweeps of the two approaches against one another here; we note only that the smaller, cheaper-per-application patches of the Schur reduction are what make the resulting solver practical at the resolutions used for the robustness study above, and that closing the iteration-count gap to the monolithic approach, for instance by tuning the damping factor or the number of patch sweeps per level, is left for future tuning.

5.7. A smaller patch: single-column smoothing. The star patch of Section 3.4 spans the two columns of cells around a vertex of the base mesh. A smaller patch is suggested by the geometry: as the aspect ratio decreases the domain becomes thinner and the divergence is dominated by its vertical derivative, so the divergence-free fields that a robust patch must contain (Section 3.4) are expected to be captured within a single column of cells. We therefore replace the vertex star on the multigrid levels by a single-column patch, the closure of one cell of the base mesh: one column of cells together with the horizontal-velocity degrees of freedom on its two vertical facets, obtained with `firedrake.ASMVankaPC` and `pc_vanka_construct_dim:1` in place of `firedrake.ASMStarPC` (Appendix B). For $k = 2$ it has $18N_z + 4$ unknowns against $30N_z + 8$ for the star (364 against 608 at $N_z = 20$).

Figures 14 and 15 show the four sweeps of Sections 5.3 and 5.4, with the same shifts $\delta = C_1/\Delta t$, meshes, aspect ratios and tolerances, for both patches. The single-column patch converges in every run in which the star patch converges, and its iteration counts are bounded in Δx, Δz, ε, Δt and C_1. For the linear problem the two patches give the same counts to within one iteration in all but the thinnest domains,

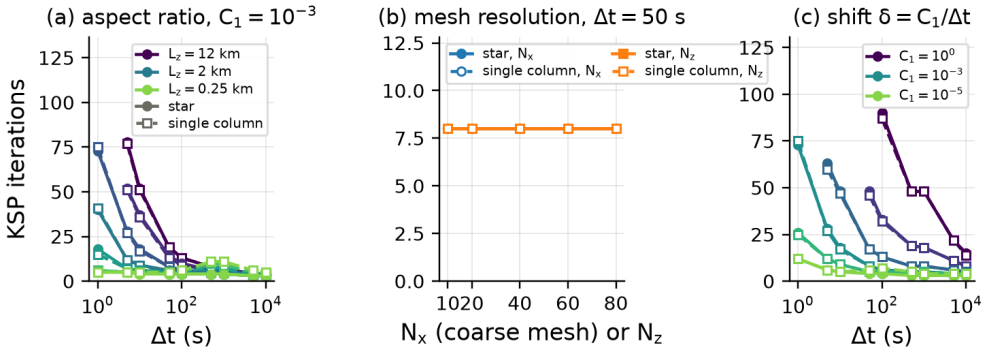


FIG. 14. *Linear problem: star patch (solid lines, filled markers) against single-column patch (dashed lines, open markers) on the sweeps of Section 5.3, $\delta = C_1/\Delta t$. (a) Aspect ratio, $C_1 = 10^{-3}$; three of the seven values of L_z are labelled, the colour runs from 12 km (dark) to 0.25 km (light). (b) Mesh resolution at $\Delta t = 50$ s, $C_1 = 10^{-3}$: the four curves coincide at 8 iterations. (c) Shift sweep on the nominal geometry; three of the six values of C_1 are labelled. Runs that reached the iteration cap are omitted.*

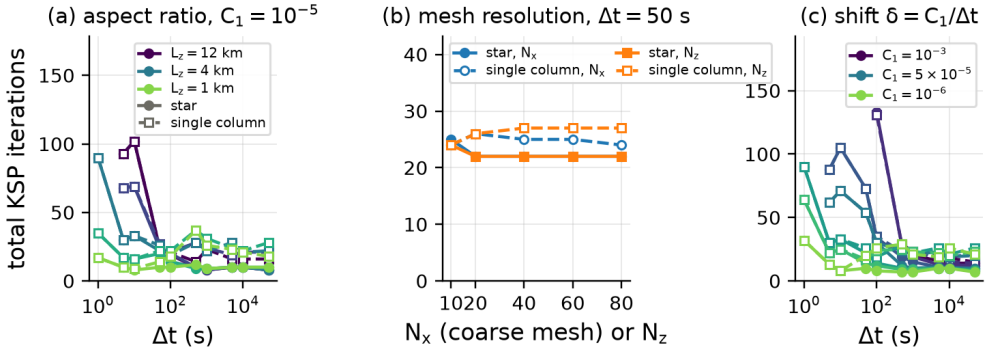


FIG. 15. *Nonlinear problem: the same comparison on the sweeps of Section 5.4 (total FGMRES iterations over the Newton steps of one time step). (a) Aspect ratio, $C_1 = 10^{-5}$; three of the five values of L_z are labelled. (b) Mesh resolution at $\Delta t = 50$ s. (c) Shift sweep on the nominal geometry; three of the seven values of C_1 are labelled.*

where the single column needs up to 2.75 times the iterations at $\Delta t = 500$–1000 s (11 against 4 at $L_z = 250$ m); the figures of Section 5.3 at $\Delta t = 50$ s are reproduced exactly for every mesh and to within one iteration for every aspect ratio. For the nonlinear problem the counts coincide where the shift is at or above the admissible threshold of Corollary 4.13 ($\Delta t \leq 100$ s for $C_1 = 10^{-5}$) and the single column needs two to three times the iterations below it ($\Delta t \geq 500$ s), with the same set of non-converging runs. The smaller patch is therefore a working, parameter-robust alternative to the star patch, at about 30% lower cost per outer iteration, so that it is also faster wherever the two counts are within a factor of about 1.4. All runs use $k = 2$; for $k = 1$ a single column contains no divergence-free fields and the star patch is required.

5.8. Discussion. The numerical results confirm parameter robustness of the preconditioner across the four axes considered, and the agreement with the shift-parameter analysis of Section 4 is quantitative rather than merely qualitative once the mode content of the test data is taken into account (Remark 5.1): the worst-case

threshold of Section 4 is sharp for the depth-independent pressure mode, which the data do not contain, and the same analysis applied to the modes they do contain locates the observed convergence boundary and explains why thinner domains converge faster although the bound is uniform in the aspect ratio.

A point worth flagging is that the analysis of Section 4 assumes an exact solve on the Schur complement block, while the implementation of Section 3 replaces this with a multigrid V-cycle with column-patch smoothing. The observed parameter robustness indicates that the multigrid V-cycle is a sufficiently accurate approximation of the exact Schur solve to preserve the iteration counts predicted by the theory. A rigorous analysis of the multigrid approximation in the presence of the grad-div augmentation (3.12) is beyond the scope of the present work; we note in particular that Arnold, Falk and Winther [1] establish the spectral equivalence of the augmented mass operator with the $H(\mathrm{div})$ Riesz map, which provides the theoretical foundation for the multigrid choice but does not by itself control the iteration count of the outer Krylov method.

6. Conclusions. We have proposed and analysed a preconditioner for the saddle-point system that arises at each step of an implicit time discretisation of the linearised Boussinesq equations on thin extruded meshes. The construction rests on a single device: augmenting the pressure equation by a small multiple δ of the pressure mass matrix. The resulting shifted matrix A_δ has an invertible pressure block, which permits an exact Schur-complement elimination of the pressure and reduces each solve to a velocity–buoyancy block carrying a grad–div augmentation. That block is treated by multigrid with a vertical column-patch line smoother, whose patches are smaller than the monolithic saddle-point patches they replace because they contain no pressure degrees of freedom. The quality of the preconditioner is then governed entirely by how well the shifted problem approximates the unshifted one, and it is this spectral distance that the analysis of Section 4 bounds.

The theoretical contribution is a bound on $\|\mathcal{A}_\delta^{-1}\mathcal{A}_0 - I\|$ with explicit dependence on δ, the time step Δt, the buoyancy frequency N, and the aspect ratio ε. It is obtained by eliminating the buoyancy exactly, reducing the problem to a perturbed mixed problem, and estimating the four Brezzi constants in a norm weighted by the stratification operator M and posed on a vertically rescaled domain. That choice of norm is what makes the estimates clean: the coercivity constant is $\alpha = 1$ exactly, independently of Δt, N, f and ε. We give two bounds: a symmetric one (Theorem 4.10), which applies to the non-rotating vertical-slice configuration used in the experiments and follows from the classical symmetric perturbed theory, and a general one (Theorem 4.11) obtained without a symmetry assumption. They agree to leading order despite coming from different sources and different arguments, which is mutual corroboration of both. They differ in that the symmetric bound holds for every $\delta > 0$, whereas the general one carries a smallness condition. In the symmetric case that condition can be shown to be unnecessary, since the shifted pressure operator is $S + \delta I$ with S positive semi-definite and the shift can only improve its conditioning.

Two consequences of the bound are worth restating. First, Corollary 4.6 evaluates C_M exactly on a box and shows it to be *exactly independent of* ε whenever $\varepsilon\,\Delta t\,N \lesssim 1$, so the inf–sup constant, and with it the whole bound, is uniform in the aspect ratio — the preconditioner is therefore well suited to precisely the limit in which standard approaches degrade. Second, Proposition 4.12 identifies two regimes for the admissible shift, $\delta \propto \Delta t$ and $\delta \propto 1/\Delta t$, meeting at $\Delta t \sim 1/(\varepsilon N)$. The controlling group is $\varepsilon\,\Delta t\,N$ rather than $\Delta t\,N$, so the aspect ratio enters the cross-over on the same footing as

the stratification. For data without a depth-independent pressure component, which includes the test problem of Section 5, the horizontal branch of C_M is never active and the change of scaling occurs instead at $\Delta t\, N \sim 1$; both the linear and the nonlinear sweeps show it there.

Numerically, the preconditioner is robust in horizontal resolution, vertical resolution, aspect ratio, and shift parameter, for the linear system and for the nonlinear problem driven by an outer Newton iteration. The predicted *scalings* are reproduced — in particular the quadratic tightening of the admissible shift as the time step decreases — and, once the mode content of the data is taken into account, so is the *threshold*: the worst-case condition of Section 4 is sharp for the depth-independent pressure mode, which the Skamarock–Klemp data do not contain, and the same analysis applied to the modes they do contain locates the observed convergence boundary to within a factor of a few (Remark 5.1). This also explains why thinner domains converge faster although the bound is uniform in the aspect ratio. A second, verification-only test case, stratified flow over a ridge on a terrain-following mesh with a mean flow (Section 5.5), shows that the solver survives a deformed geometry, and supplies an independent check on the theory. Two implementation findings are of practical importance. Reverse Cuthill–McKee reordering of the patch subproblems is a necessity. Without it the patch bandwidth grows with the number of vertical layers and the preconditioner application cost grows superlinearly, while with it the cost per layer is constant. And the comparison against a monolithic column-patch smoother applied directly to A_δ is, in the present untuned configuration, not in favour of the Schur reduction. The fieldsplit patches are about 20% smaller and cheaper per iteration, but the monolithic patches resolve more of the pressure–velocity coupling per sweep and converge in fewer FGMRES iterations.

Several limitations should be stated plainly. The analysis assumes an exact solve on the Schur-complement block, whereas the implementation uses a multigrid V-cycle; the observed robustness indicates that the V-cycle is an adequate substitute, but this is verified numerically rather than proved. The numerical experiments are two-dimensional vertical slices, a simplification, although the analysis is carried out for the general three-dimensional system. The worst-case cross-over of Proposition 4.12 at $\Delta t \sim 1/(\varepsilon N)$ has not been exhibited: it is visible only to data with a depth-independent pressure component, which neither test case contains, and it lies beyond the time steps tested (Remark 5.1); what the experiments do show is the change of scaling on the vertical branch, at $\Delta t\, N \sim 2$. Finally, the worst-case bound is a guarantee for general data, and is sharp only for the depth-independent mode; a sharp threshold for a given right-hand side requires knowing its mode content.

Each of these limitations suggests a corresponding next step, and Section 7 sets out the directions we intend to pursue.

7. Future work. [Review: Written for the annual progress review; not intended for the journal version, where the closing paragraph of Section 6 would carry an abbreviated form of it. Condense or remove before submission, together with the pointer to this section in the Introduction's roadmap paragraph.]

The directions below fall into three groups: completing the analysis and the evidence for the solver as it stands, extending the discretisation it is built on, and applying it to problems that a less efficient solver would put out of reach.

7.1. Completing the analysis and the evidence.

The most conspicuous gap between the theory and the implementation is that Theorems 4.10 and 4.11 assume an *exact* solve on the Schur-complement block S, whereas the solver of Section 3.4

replaces it with a single multigrid V-cycle using a vertical column-patch smoother. The numerical evidence of Section 5.3 indicates that the substitution costs nothing in iteration count, but this is not proved. What is available is the spectral equivalence of the grad–div augmented mass operator with the $H(\mathrm{div})$ Riesz map [1], which justifies the choice of multigrid without controlling the outer Krylov iteration count. Section 3.4 supplies the horizontal half of the kernel argument — the divergence-free basis, obtained as $\nabla^{\perp}$ of the continuous CG_k basis, is supported on base-mesh vertex stars — but the vertical (column) part of the argument, in the anisotropic $\varepsilon \to 0$ limit, remains open. The objective is a parameter-robust two-level or multilevel bound for the patch smoother, uniform in Δt, N, ε and the mesh, with the small line-patch subproblem as the local solve. The difficulty is the anisotropic thin-cell limit: the column patch is chosen precisely because it captures the strong vertical coupling that small ε induces, so any useful bound has to track that coupling rather than assume it away, and the grad–div term of (3.12) must be carried through the local analysis. A successful argument here would close the last gap between the bound of Section 4 and the algorithm actually run.

The ridge case of Section 5.5 establishes that the solver works on a deformed, terrain-following mesh with a mean flow, but it is a verification exercise only: the four robustness sweeps of Section 5.3 have not been repeated on it. Doing so is the immediate next experiment, and it is more than bookkeeping, since the ridge configuration sits at $\varepsilon \approx 0.24$, an order of magnitude fatter than the nominal slice, and so probes a part of the parameter space the existing sweeps do not. Beyond that, both test cases are deliberately weakly nonlinear — the ridge case has $Nh_m/U = 10^{-3}$ and stays within the linear Queney regime. Raising Nh_m/U until the flow steepens and the wave field breaks would test the coupling between the outer Newton iteration and the shifted preconditioner in a regime where the Jacobian genuinely varies between steps, which is the situation the preconditioner was designed for has not yet been stressed by.

The analysis of Section 4 is carried out for the general three-dimensional system, but every experiment reported here is a two-dimensional vertical slice. Three-dimensional experiments are therefore the natural next step; there, the column patches acquire a larger horizontal footprint. Corollary 4.6 implicitly assumes the two horizontal extents are comparable; for $L_y \neq L_x$ the extremal horizontal mode changes and the constant becomes $C_P = (2\pi)^{-1}\max(1, L_y/L_x)$, so a strongly anisotropic horizontal geometry needs to be checked.

7.2. Extending the discretisation. The time discretisation used here is the implicit midpoint rule, the one-stage Gauss–Legendre collocation method. It is natural to start with, since it is second-order, symplectic, and produces a single saddle-point system per step of exactly the form analysed, but it is also the simplest member of the family that Irksome makes available. Higher-order or multi-stage implicit Runge–Kutta schemes raise a structural question for the preconditioner rather than merely a computational one: an s-stage method couples the stages, so the system to be solved at each step is an $s \times s$ block system whose diagonal blocks each have the saddle-point structure treated here. Whether the shift-and-eliminate strategy should be applied stagewise, to each diagonal block within an outer stage iteration, or to the coupled stage system as a whole, is open. This matters most for problems where the accuracy of the time discretisation, rather than its stability, is the binding constraint.

The ideal Boussinesq system is Hamiltonian with respect to a non-canonical Poisson bracket, with the total energy as Hamiltonian. The target here is a discretisation

that preserves that Poisson structure. Energy conservation follows from antisymmetry of the bracket alone, whereas the bracket is also degenerate, and its degeneracy supplies Casimir invariants that are conserved for *any* Hamiltonian. A scheme that inherits the bracket therefore inherits a whole family of invariants rather than a single one, and constrains the solution over the entire integration rather than over each step, which matters for integrations of the length ocean models perform. Compatible finite element spaces are a natural setting for this, since the discrete de Rham structure already supplies the operators the bracket is built from [7]. It is worth noting where the present scheme stands: the implicit midpoint rule is symplectic and preserves quadratic invariants, but is not in general a Poisson integrator for a non-canonical bracket, so what is used here is a partial result rather than the goal.

The solver connection is that such schemes are implicit of necessity and typically introduce additional nonlinear coupling, which is the usual reason they are set aside as too expensive. That objection is what an efficient implicit solver removes.

7.3. Applications. A more speculative but attractive target is the Lagrangian-averaged, or α, model built on the incompressible Boussinesq system. In this family the transported velocity is related to the smoothed velocity by a Helmholtz filter, $\vec{v} = (I - \alpha^2\Delta)\vec{u}$, which regularises the nonlinearity at scales below α while retaining a Kelvin circulation theorem and a Hamiltonian structure of the same Lie–Poisson type as the unregularised system, differing in the metric that defines the energy. It is therefore a turbulence closure with genuine geometric structure rather than an eddy-viscosity fit, which makes it a good match for a compatible discretisation and for the structure-preserving direction above, to which it is more closely related than it first appears: the two share the same geometric setting, so a Poisson integrator developed for the Boussinesq bracket should carry over with the metric replaced. The computational obstacle has always been the extra elliptic solve that the Helmholtz filter requires at every step, on top of the implicit system. That solve is of precisely the type the machinery developed here already handles, which is an $H(\mathrm{div})$ problem on the same extruded, high-aspect-ratio mesh, amenable to the same column-patch multigrid, so the marginal cost of the α model over the Boussinesq system may be far smaller in this framework than it appears in general. Establishing that, and then using it to study stratified turbulence at resolutions where the unregularised system is intractable, is the longer-term aim.

Appendix A. Time discretisation in Irksome.

The fully implicit Runge–Kutta time discretisation of Section 2.3 is implemented with the Irksome package [14], which operates directly on the symbolic semi-discrete variational form (2.16): each time derivative is replaced by a symbolic operator `Dt`$(\cdot)$, and, given a Butcher tableau, Irksome assembles the coupled stage residual and Jacobian automatically from the per-stage spatial operators. The stage-coupled system is solved with the preconditioner of Section 3, supplied to PETSc as the preconditioning operator for each stage block; since the per-stage operator has exactly the block structure of (3.2), the fieldsplit configuration used for the one-stage implicit midpoint rule in Section 5 should extend unchanged to higher-stage methods. Concretely, the time stepper is constructed as

```
butcher_tableau = GaussLegendre(1)     # implicit midpoint rule
stepper = TimeStepper(eqn, butcher_tableau, t, dt, U,
                      bcs=bcs, solver_parameters=params, appctx=appctx)
```

where `eqn` is the semi-discrete residual written with `Dt`$(\cdot)$, and `t`, `dt` are `MeshConstant` objects so that the time step can be varied without re-generating code. Four aspects

of the configuration are not incidental to the method.

First, the outer solve is matrix-free (`mat_type:matfree`): the stage-coupled operator is never assembled, only the auxiliary operator of Appendix C, which is the object the preconditioner needs; this is what makes the shift strategy affordable.

Second, the preconditioner is a subclass of Irksome's `IRKAuxiliaryOperatorPC` that overrides `getNewForm` to return the auxiliary form of Appendix C.1; this is the mechanism by which the shifted operator A_δ, rather than A_0, is presented to PETSc as the preconditioning operator. The shift itself is not hard-coded but passed through Irksome's application context, `appctx = {"dt": dt, "shift": delta, ...}`, and read back inside the preconditioner by `self.get_appctx(pc)`.

Third, the nonlinear experiments of Section 5.4 use Newton's method with the Eisenstat–Walker adaptive forcing term (`snes_type:newtonls`, `snes_ksp_ew`, initial relative tolerance 10^{-2}), so that the inner Krylov tolerance is tightened only as the Newton iteration approaches the solution. This requires right preconditioning, `ksp_pc_side:right`, and is why the convergence histories of Section 5.4 are reported against *cumulative* inner iterations: the per-step iteration count is not constant by design.

Finally, the constant pressure mode, the nullspace of the unshifted operator that the analysis removes by taking $Q = L_0^2$ in Section 4.2, is declared to the solver as a `MixedVectorSpaceBasis` whose pressure component is the constant-function basis.

Appendix B. Firedrake configuration of the column-patch smoother.

The column-patch smoother of Section 3.4 is realised through `firedrake.ASMStarPC` with `pc_star_construct_dim:0`, which constructs the star patch at base-mesh vertices and automatically extrudes it through the column, giving exactly the patches ω_i and local spaces X_i of Section 3.4. The velocity–buoyancy block, which is the second fieldsplit in the ordering of Appendix C.1, is configured as

```
"fieldsplit_1": {
    "ksp_type": "preonly",
    "pc_type": "mg",
    "pc_mg_type": "full",
    "pc_mg_cycle_type": "v",
    "mg_levels": {
        "ksp_type": "richardson",
        "ksp_max_it": 6,
        "pc_type": "python",
        "pc_python_type": "firedrake.ASMStarPC",
        "pc_star_construct_dim": 0,
        "pc_star_sub_sub_pc_type": "lu",
        "pc_star_sub_sub_pc_factor_mat_ordering_type": "rcm",
        "pc_star_sub_sub_pc_factor_reuse_ordering": None,
    },
    "mg_coarse": {
        "ksp_type": "preonly",
        "pc_type": "lu",
    },
}
```

The entries divide into three groups.

The patch construction. The extrusion of the star patch through the whole column is what makes this a line smoother rather than a point smoother; `pc_star_sub_sub_`

`pc_type:lu` then performs the direct factorisation of each local patch operator A_i.

The reordering. The two `pc_star_sub_sub_pc_factor` options are what realise the timing result of Section 5.6: `mat_ordering_type:rcm` applies reverse Cuthill–McKee to each patch before factorisation, and `reuse_ordering` computes the permutation once and reuses it, since the patch sparsity pattern does not change between time steps or Newton steps. Without the first, the patch bandwidth grows with the number of vertical layers and the local solve degenerates from a line solve into a dense factorisation.

The relaxation. `ksp_type:richardson` with `ksp_max_it:6` applies six Richardson sweeps per multigrid visit, with the damping factor θ. Neither has been tuned to be optimised, which is the sense in which the comparison of Section 5.6 between the fieldsplit and monolithic smoothers is untuned on both sides. The coarse grid is solved directly.

The pressure block, by contrast, requires no iteration at all:

```
"fieldsplit_0": {
    "ksp_type": "preonly",
    "ksp_reuse_preconditioner": None,
    "pc_type": "python",
    "pc_python_type": "firedrake.AssembledPC",
    "assembled_pc_type": "bjacobi",
    "assembled_sub_pc_type": "ilu",
}
```

The reasoning behind this choice, and the sense in which it is exact, is given in Appendix C.4.

Appendix C. The auxiliary-form Schur complement.

This appendix describes how the exact Schur complement (3.13) is realised inside the Irksome/PETSc solver pipeline, and establishes that the object PETSc assembles is that Schur complement rather than an approximation to it. Appendix C.1 constructs the auxiliary form and states the result; Appendices C.2–C.4 prove it by tracking how PETSc builds the fieldsplit preconditioner from a user-supplied operator, and identifying the role played by each block.

C.1. Auxiliary operator construction in Irksome.

The Schur-complement strategy of Section 3.3 is conceptually straightforward, but realising the exact Schur complement (3.13) inside the solver pipeline used for the numerical experiments requires presenting the solver with an *auxiliary form* $\hat{F}$ that differs from the residual of A_δ by the grad-div augmentation (3.12) added to the velocity equation. Symbolically, $\hat{F}$ is obtained from the residual F of A_δ by the single substitution

$$p'_h \;\longmapsto\; p'_h - \frac{1}{\delta}\,\nabla\cdot\vec{u}'_h, \tag{C.1}$$

applied wherever the pressure appears, with the shifted divergence equation $\langle\phi, \nabla\cdot\vec{u}'_h\rangle + \delta\langle\phi, p'_h\rangle = 0$ retained as in A_δ. Substituting (C.1) into the pressure-gradient term of the momentum equation gives

$$-\langle\nabla\cdot\vec{w},\, p'_h\rangle \;\longmapsto\; -\langle\nabla\cdot\vec{w},\, p'_h\rangle + \frac{1}{\delta}\langle\nabla\cdot\vec{w},\, \nabla\cdot\vec{u}'_h\rangle, \tag{C.2}$$

which is exactly the grad-div augmentation (3.12).

Two points about (C.1) are worth making. It is implemented literally, as a substitution on the symbolic pressure variable before the residual is built, so the auxiliary

form is written once and the grad-div term is generated automatically in every equation where the pressure occurs. And no mass-matrix inverse appears in it, although M_p^{-1} appears in the matrix expression (3.6): the two agree because the compatible triple (2.14) satisfies $\nabla \cdot H_h(\mathrm{div}) \subseteq L_h^2$ exactly, so the L^2 projection of $\nabla \cdot \vec{u}'_h$ into the pressure space is $\nabla \cdot \vec{u}'_h$ itself and M_p^{-1} acts as the identity at the level of forms. This is the same structural property that makes the buoyancy elimination of Remark 4.1 exact, and it is what allows the auxiliary form to be written without ever inverting a mass matrix symbolically.

PROPOSITION C.1 (Schur complement via auxiliary form). *Let $\hat{F}$ be the auxiliary form defined above. If a Schur-complement fieldsplit method eliminates the pressure variable first and takes its preconditioning operator for the remaining velocity–buoyancy block from the $(1,1)$ block of the operator assembled from $\hat{F}$, then the resulting preconditioner is the exact Schur-complement preconditioner for A_δ, with velocity–buoyancy block preconditioner equal to the Schur complement* (3.13).

In the implementation $\hat{F}$ is supplied to PETSc through the `IRKAuxiliaryOperatorPC` subclass of Appendix A, with the fieldsplit options

```
"pc_fieldsplit_type": "schur",
"pc_fieldsplit_schur_fact_type": "full",
"pc_fieldsplit_schur_precondition": "a11",
"pc_fieldsplit_0_fields": "3",       # pressure
"pc_fieldsplit_1_fields": "0,1,2",   # velocity, buoyancy
```

The field ordering places the pressure in the first split, so that it is the field eliminated first and the Schur complement is formed in the velocity–buoyancy space. The remainder of this appendix proves the proposition.

C.2. The PETSc fieldsplit Schur factorisation. Following the field ordering declared above, we place the pressure p' first and the velocity–buoyancy field $\vec{y} = (\vec{u}', b')$ second, and write the shifted operator in the two-by-two block form

$$A_\delta = \begin{pmatrix} \delta M_p & E \\ B & D \end{pmatrix}, \tag{C.3}$$

where δM_p is the shifted pressure block, D is the velocity–buoyancy transport block, and B, E are the pressure couplings; this is (3.9) with the blocks permuted into PETSc's split order. For an operator with this structure, a Schur fieldsplit with the `full` factorisation type forms the preconditioner from

$$A_\delta^{-1} = \begin{pmatrix} I & -(\delta M_p)^{-1} E \\ 0 & I \end{pmatrix} \begin{pmatrix} (\delta M_p)^{-1} & 0 \\ 0 & S^{-1} \end{pmatrix} \begin{pmatrix} I & 0 \\ -B(\delta M_p)^{-1} & I \end{pmatrix}, \tag{C.4}$$

in which

$$S = D - B\,(\delta M_p)^{-1}\,E \tag{C.5}$$

is the Schur complement with respect to the pressure block, an operator on the velocity–buoyancy space; it is (3.11). The pressure is thus the field eliminated first, which is the arrangement the analysis of Section 3.3 assumes.

In practice PETSc does not form $(\delta M_p)^{-1}$ or S^{-1} directly. Each appears through an inner solve, which we denote $\mathrm{ksp}(\delta M_p, P_p)$ for the pressure block preconditioned by an operator P_p, and $\mathrm{ksp}(\hat{S}, \hat{S}_p)$ for the Schur complement preconditioned by an

operator $\hat{S}_p$. The assembled preconditioner is therefore

$$\begin{pmatrix} \mathrm{ksp}(\delta M_p, P_p) & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} I & -E \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ 0 & \mathrm{ksp}(\hat{S}, \hat{S}_p) \end{pmatrix} \begin{pmatrix} I & 0 \\ -B\,\mathrm{ksp}(\delta M_p, P_p) & I \end{pmatrix}. \tag{C.6}$$

Its accuracy is governed by two choices, the preconditioning operator $\hat{S}_p$ for the Schur complement and the preconditioning operator P_p for the pressure block, which are treated in turn in Appendices C.3 and C.4. The off-diagonal blocks B and E are taken directly from the operator supplied to the preconditioner and require no approximation: they agree with those of A_δ, since $\hat{F}$ and the residual of A_δ differ only in the diagonal velocity–buoyancy block.

C.3. The role of the auxiliary form. The Schur complement (C.5) is dense and is not assembled explicitly. PETSc instead constructs the approximation $\hat{S}_p$ from a user-supplied operator, according to the option `pc_fieldsplit_schur_precondition`; the setting `a11` instructs it to take $\hat{S}_p$ to be the $(1,1)$ block of that operator, the block belonging to the second split — here the velocity–buoyancy field, since the pressure was placed first.

This is where the auxiliary form enters. Under the substitution (C.1), the momentum equation of $\hat{F}$ acquires the grad-div term (C.2), so the velocity–buoyancy block of $\hat{F}$ is not D but

$$D + \frac{\Delta t}{\delta} G M_p^{-1} G^\top, \tag{C.7}$$

which is precisely the exact Schur complement (3.13): the substitution (C.1) is the elimination of the pressure through the shifted divergence equation performed in Section 3.3, carried out at the level of forms, and (C.2) is the resulting grad-div correction, with the factor Δt of the momentum equation. When PETSc reads the $(1,1)$ block of the operator assembled from $\hat{F}$ under the `a11` option it therefore obtains $\hat{S}_p = S$ exactly, rather than an approximation to it.

C.4. Exactness of the pressure block solve. It remains to confirm that the inner solve on the pressure block introduces no approximation either. That block is δM_p, where M_p is the mass matrix on the discontinuous pressure space L^2_h. Because L^2_h has no inter-element continuity, M_p is block-diagonal with one dense block per element. The configuration of Appendix B exploits this: block-Jacobi (`bjacobi`) splits the matrix along exactly those element blocks, and the incomplete factorisation applied to each block is in fact complete, since a dense block has no fill-in to discard, so `ilu` coincides with `lu` there. The wrapper `firedrake.AssembledPC` is required only because these preconditioners need an assembled matrix, whereas the outer solve is matrix-free (Appendix A). Finally `ksp_type:preonly` applies the resulting factorised inverse exactly once with no Krylov iteration, and `ksp_reuse_preconditioner` avoids recomputing a factorisation that does not change between steps. The pressure solve is therefore exact up to rounding.

Consequently the only approximation in (C.6) is the inner solve $\mathrm{ksp}(\hat{S}, \hat{S}_p)$ on the Schur complement, and by Appendix C.3 that solve preconditions the exact Schur complement by itself. In the experiments reported here it is applied with `ksp_type:preonly`, so that a single multigrid V-cycle of Appendix B is used in place of S^{-1}; this is the one place where the implemented preconditioner departs from the operator analysed in Section 4, as noted in Section 5.8.

Taken together, Appendices C.2–C.4 establish Proposition C.1: the block PETSc extracts from $\hat{F}$ under the `a11` option is exactly S, the off-diagonal couplings agree

with those of A_δ, and the pressure solve is exact, so the preconditioner assembled from $\hat{F}$ is the exact Schur-complement preconditioner for A_δ, without any explicit assembly of the dense Schur complement.

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